Asymmetric Labor Income Risk and Stock Market Participation: Evidence from the US

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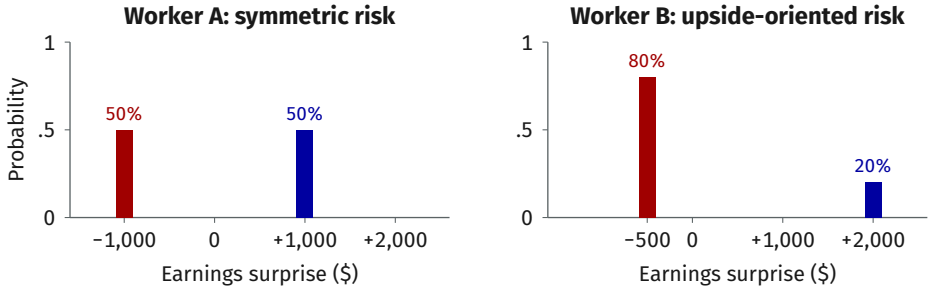
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Overview

- Labor income is a major source of household risk. Earnings gains and losses can have different implications for stock-market participation.
- Does the balance of upside and downside earnings variation predict entry, *holding total variance fixed*?
- This paper:
  1. Relates earlier earnings distributions across worker groups to subsequent entry among initial nonholders.
  2. Uses a saving model with participation costs to examine the roles of realized earnings and future earnings risk.

A given variance can reflect large gains, large losses, or both. These differences can affect the resources available for investment.

Earnings distributions with the same mean and variance



Same mean surprise: \$0 Same variance: $\$1{,}000^2$

Distribution B has smaller frequent losses and larger rare gains.

Preview of results

1. Earnings upside balance predicts stock-market entry, conditional on mean growth, variance and household resources.
   - A one-SD increase in upside balance predicts **1.60 percentage points more annual entry**.
   - Mean entry: 8.82%; 27,419 SIPP observations of initial nonholders.
2. The association also appears for entry followed by later ownership.
3. The model reproduces the positive sign but understates the magnitude.
   - Realized earnings help finance entry.
   - Future earnings risk changes saving and subsequent participation.

▶ Contribution ▶ Economic magnitude ▶ Published benchmarks

Related literature

- Earnings risk and portfolio choice: d'Astous & Shore (2024); Bacher (2025).
  - Earnings risk, household resources and portfolio choice.
- Downside earnings exposure and stock-market risk: Catherine, Sodini & Zhang (2024); Shen (2024).
  - Countercyclical downside earnings risk and household portfolios.
- Career earnings dynamics: Gálvez & Paz-Pardo (2026).
  - Richer earnings processes change participation and inferred preferences.

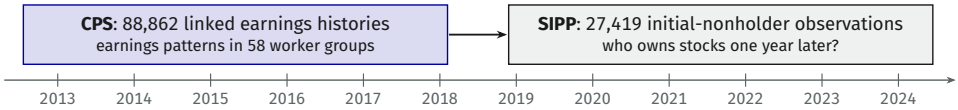
This paper relates earnings upside balance to entry at similar mean growth and variance, then evaluates the relationship in a model fitted to other outcomes.

▶ Earlier evidence ▶ Closest studies ▶ Related work

Data and earnings groups

- **Current Population Survey (CPS):** earlier annual wage changes, including workers whose wages fall to zero the next year.
- **Survey of Income and Program Participation (SIPP):** later household assets and stock-market entry.

58 earnings cells with sufficient data: industry × education × age band.
Education: high school or less / some college / college. Age: 25–44 / 45–65.



Entry: no stocks or mutual funds today, ownership next year, outside retirement accounts.

▶ How earnings are measured ▶ Samples ▶ Timing

Measuring upside and downside earnings variation

Compare earnings distributions with the same total variance. More upside means shifting variation from below-average to above-average growth.

For age-adjusted wage growth ϵ in cell c, with mean μ_c:

$$V_c^+ = \mathbb{E}_c[(\epsilon - \mu_c)^2 \mathbf{1}\{\epsilon > \mu_c\}], \qquad V_c^- = \mathbb{E}_c[(\epsilon - \mu_c)^2 \mathbf{1}\{\epsilon < \mu_c\}].$$

Total variance

$$V_c = V_c^+ + V_c^-$$

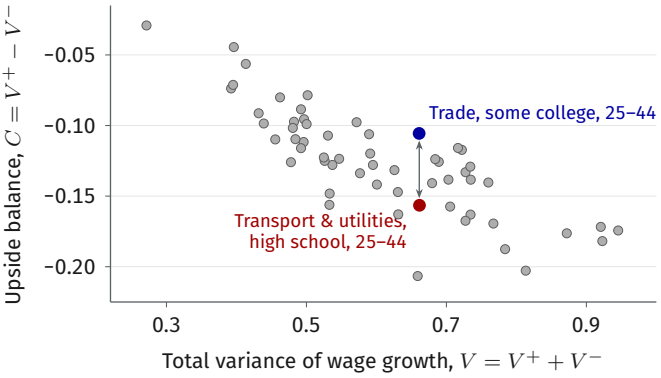
Upside balance

$$C_c = V_c^+ - V_c^-$$

The regression holds V fixed: higher C means more upper and less lower semivariance.

▶ Construction ▶ Numerical example ▶ Skewness

Variation in earnings risk across worker groups



58 worker groups

All have $C < 0$: below-average changes account for more of the variance.

$\text{corr}(C, V) = -0.81$.

Cells with more transitions to zero annual wages have higher V and lower C.

▶ Economic drivers ▶ Highlighted pair ▶ Cell definitions

Empirical specification

Compare initial nonholders facing different earnings distributions, conditional on mean growth, total variance and observed household characteristics.

$$\underbrace{\mathbf{1}\{\text{Participates}_{i,t+1}\}}_{\text{entry next year}} = \alpha + \beta \frac{C_{c(i,t)} - \bar{C}}{s_C} + \text{controls}.$$

Comparison	Controls
Earnings risk	Variance V and standard deviation $\sqrt{V}$
Mean earnings growth	Mean growth and its square
Current resources	Assets, earnings, nonpositive-earnings indicator
Demographics	Age, age squared, sex
Fixed effects	Industry, education, age band, year

Sample: initial nonholders only. Stocks and mutual funds outside retirement accounts. Weighted linear probability model; SEs clustered by industry–education group.

▶ Full specification ▶ Construction versus controls ▶ Person FE

Economic magnitude

A one-SD increase in upside balance predicts 1.60 pp more annual entry (SE 0.51).

16 additional entrants per 1,000 initial nonholders; 18% of mean entry (8.82%).

Published one-SD comparisons:

Study	Predictor and outcome	Relative change
This paper	Upside balance; annual entry	$+18\%$
Kaustia–Knüpfer (2012)	Local-peer returns; next-month entry	$+9$–13%
d'Astous–Shore (2024)	Permanent earnings volatility; ownership	-23%

These studies also find substantial relative participation responses. Their predictors, populations and horizons differ.

▶ Sources and absolute magnitudes ▶ Regression estimates ▶ Units

Sensitivity to the variance specification

Dependent variable: entry next year	(1)	(2)	(3)	(4)
Upside balance C	—	—	1.111^{**}	1.601^{***}
			(0.526)	(0.506)
Variance V	-0.457	-11.839^{***}	0.053	-15.380^{***}
	(0.584)	(3.845)	(0.548)	(3.505)
Standard deviation $\sqrt{V}$	—	11.457^{***}	—	15.762^{***}
		(3.689)		(3.456)
Mean-growth controls	Yes	Yes	Yes	Yes
Resource and demographic controls	Yes	Yes	Yes	Yes
Industry, education, age-band, year FE	Yes	Yes	Yes	Yes
Observations	27,419	27,419	27,419	27,419

Notes: Survey-weighted estimates, in pp per SD. Each earnings measure uses its CPS SD. Parentheses: SEs clustered by industry–education (30 groups); t_{29} tests. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Columns (2) and (4) allow curvature in variance; (3) is equivalent to entering V^{+}, V^{-} directly.

The upside-balance coefficient remains positive with a linear variance control.

▶ Direct upper/lower coefficients ▶ Why opposite variance signs?

Additional evidence

Comparison	Evidence and interpretation
Measurement and sorting	Positive entry estimates under alternative earnings measures, richer group controls and reporting screens.
External comparison	Panel Study of Income Dynamics (PSID): no clear association over ages 25–65; a positive estimate at 25–44 comes from an exploratory split.
Subsequent ownership	SIPP: the association also appears for entry followed by later ownership.
Reported expectations	New York Fed SCE: directly reported earnings expectations and changes in retirement portfolios.

The surveys answer different questions: entry, external comparison, and subjective expectations.

▶ Controls and reporting ▶ PSID ▶ Samples ▶ Earnings measures

Entry followed by later ownership

Each column compares two outcomes for the same initial nonholders.

Follow-up cohort	One later year	Two later years
Entry alone (pp)	2.09^{***} (0.70)	3.31^{**} (1.43)
Entry + later ownership (pp)	0.98^{*} (0.50)	1.89^{**} (0.79)
Sustained outcome $= 1$	$0, 1, 1$	$0, 1, 1, 1$
Initial nonholder observations	9,236	3,254

The sample includes all initial nonholders with the required annual follow-up.

▶ Entry and interruption ▶ Sample and sequences

Reported expectations and portfolio adjustment

The preceding results use historical earnings distributions. The New York Fed SCE relates reported earnings expectations to retirement-portfolio adjustment.

Belief measure	Reallocation	Equity share
Upside-minus-downside tilt	0.028^{**}	-0.576
	(0.013)	(0.664)
Sum of semideviations	-0.017	-3.805^{***}
	(0.018)	(1.011)
Respondents	2,136	3,332

More expected upside is associated with fewer moves toward safer assets. The equity-share association is imprecise.

Reallocation: $+1$ riskier, -1 safer, 0 unchanged. Equity share: percentage points. Controls: expected growth, age, education, income, state and module year; 50 state clusters. Belief and action windows overlap. This is portfolio adjustment, not market entry.

▶ Different outcomes and samples ▶ Inference

Model: earnings risk and costly participation

- A participation fee makes stock investment unattractive at low resources. Earnings gains can make participation worthwhile.
- Future earnings risk affects consumption and saving, and therefore the resources available for subsequent entry.
- Estimate earnings processes from earnings data and financial parameters from participation and wealth outcomes.
- Test whether the model reproduces the entry–risk relationship, which is *excluded from estimation*.

The 58 earnings cells provide measured differences in earnings prospects. Preferences and participation costs are common across cells.

▶ Related models ▶ Estimation sequence ▶ Alternative access model

A participation threshold with a fixed fee

1. Today: income Y arrives. Add inherited resources w to get cash $m = w + Y$.
2. Today: choose consumption and saving. Pay a fixed fee κ to invest part of saving in stocks.
3. Tomorrow: consume the return on saving. Stocks offer higher expected returns, with risk.

Optimal entry occurs above a cash threshold m^*.

$$\Pr(\text{entry}) = \Pr\left(Y \geq \underbrace{m^* - w}_{\text{income needed to reach the threshold}}\right).$$

The fixed fee absorbs a smaller share of resources when cash is higher.

▶ Equal-variance example ▶ Saving with future earnings ▶ Derivation

Resources, expectations and household choices

Realized resources	Earnings and asset returns have arrived. Cash, earnings state, investment access and the fee offer are known.
Future earnings	The household knows the distribution, and weighs consumption it could afford in each future outcome.
Current choices	Consume, save, and decide whether to hold stocks and what share to invest.

$$c_t + A_t + D_t K_t = M_t, \qquad M_{t+1} = A_t R_{t+1}(\alpha_t) + Y_{t+1}.$$

M: cash; c: consumption; A: saving; D: stockholding indicator; K: fee.

α: stock share; R: gross portfolio return; Y: disposable income.

Expectations change today's saving; saving changes tomorrow's resources.

▶ Resources, expectations and saving ▶ Numerical example ▶ Optimization

Earnings processes across worker groups

Workers share broad career patterns but differ in earnings growth, wage-work exit, and the size and balance of earnings shocks.

Shared foundation	College and noncollege career processes
30 career groups	Industry $\times$ education, with group wage levels
58 measured cells	Adjustments before and after age 45 within those groups

Each cell has four adjustments, estimated jointly from its earnings moments.

Preferences, participation costs and opportunity probability are common.

Two group–age combinations lack sufficient data and use the other age band's adjustment.

▶ Four fitted adjustments ▶ How tails differ ▶ Why 58, not 580?

The earnings process

Cell adjustments enter predictable growth and both shock distributions:

$$\log W_{gt} = \underbrace{f_h(t) + \ell_g + B_g(t; b_{g0}, b_{g1})}_{\text{predictable profile}} + \eta_t + \varepsilon_{gt}, \qquad W_{gt} > 0.$$

f_h: education age profile; ℓ_g: group wage level; b: cell growth.

Component	Distribution and cell parameters
Persistent earnings	$\eta_{t+1} \mid \eta_t \sim P_{h,t}(\eta_t, \cdot\,; a_c)$ a_c changes promotion and loss probabilities.
Temporary earnings	$\varepsilon_{gt} \sim F_h(\cdot\,; s_c, a_c)$ s_c scales the spread; a_c changes tail probabilities.
Wage-work exit	$\Pr(\text{exit} \mid \eta_t) = \min\{\delta_{h,t}(\eta_t) e^{a_c^{\text{exit}}}, 0.6\}$

Pooling: set $(b_c, s_c, a_c, a_c^{\text{exit}}) = (0, 0, 0, 0)$ **in every cell.**

$B_g = 0$; the baseline processes $P_{h,t}(\cdot; 0)$ and $F_h(\cdot; 0, 0)$ and baseline departure risk remain.

▶ What setting zero does ▶ Jump probabilities ▶ Temporary shocks

Structural estimates and external inputs

Inputs held fixed

Input	Value / source
Safe gross real return	1.02
Ordinary stock return: mean / SD	1.06 / 0.18
Annual crash probability	1%
Crash net real return	-48.5%
EIS ψ	0.5
Earnings / resources	CPS / PSID
Observed initial states	SIPP

Estimated financial parameters

Parameter	Estimate
Risk aversion γ	8.801
Discount factor β	0.811
Mean fee offer κ_R	**\$524**
Log-cost standard deviation σ_κ	0.944
Opportunity probability q	**0.176**

Simulated method of moments: five parameters, eight financial targets.

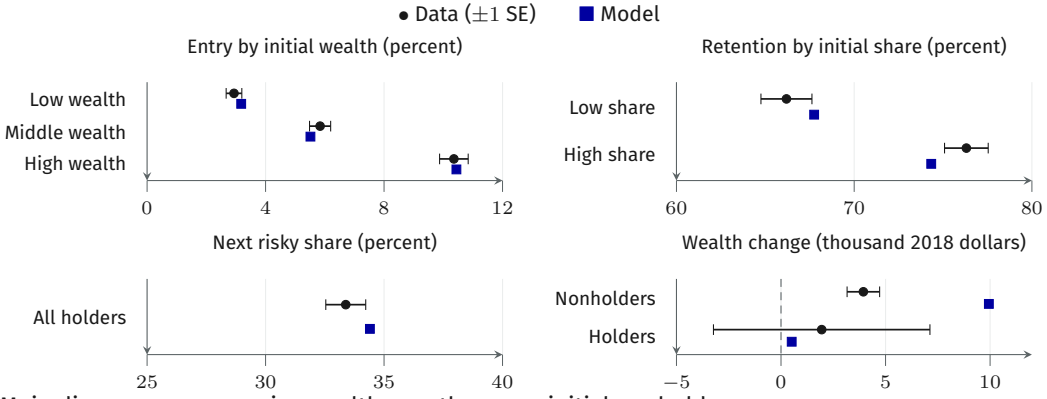
Calibrated: returns and EIS. **Estimated separately:** earnings processes.

From data: resource profiles and initial states. **Untargeted:** entry–risk coefficients.

▶ Targets and weighting ▶ Financial fit ▶ Opportunities and fees

Fit to targeted financial outcomes

Eight financial moments; five estimated parameters



Main discrepancy: excessive wealth growth among initial nonholders.

The following comparison evaluates the untargeted entry–risk relationship.

◀ Back ▶ Full target table ▶ Targets and weighting

Why compare cell and pooled earnings processes?

Do measured earnings differences across worker groups help explain why workers with more upside enter more often?

Component	Cell	Pooled
Education age profile $f_h(t)$	Same	
Group wage level ℓ_g	Same	
Cell growth profile $B_g(t)$	Fitted	Zero
Departure, scale and tail adjustments	Fitted	Zero
Asymmetric base shocks	Present in both	
Initial states and financial parameters	Same	

Both allow asymmetric earnings. Pooling removes the measured cell differences.

Same starting households and financial parameters; choices adjust to each process.

▶ Untargeted comparison ▶ Same regression ▶ What remains under pooling

Measured earnings differences improve the entry prediction

Upside-balance coefficient: percentage points per SD

Same 17,606 initial nonholders	Entry association
Observed data	1.816^{***} (SE 0.480)
Model with pooled earnings processes	-0.005
Model with cell earnings processes	$+0.241$

Cell processes add **0.247 pp** with financial parameters fixed.

The predicted relationship moves toward the data, but remains much too small.

Same entry regression, controls and weights. These coefficients were not targeted.

▶ Resources and decision rules ▶ Matched comparison ▶ Full comparison

Realized earnings account for most of the improvement

Different earnings can finance entry even when workers start with the same resources.

Keep the rule fixed to compare earnings outcomes; keep earnings outcomes fixed to compare rules.

Contribution to the upside-balance coefficient	pp per SD
Resources and earnings states reached	$+0.274$
Different optimal decision rules	-0.027
Total: cell minus pooled	$+0.247$

Changing decision rules slightly offsets the one-year gain.

To examine saving over time, I next change earnings composition within the cell model.

◀ Back ▶ Four comparisons ▶ What is a decision rule?

Changing earnings composition within the cell model

The cell–pooled comparison changes several earnings features together. Here I shift variation from below-average to above-average earnings, holding expected disposable earnings and total dollar variance fixed.

Let U_y and L_y denote upper and lower partial variances of dollar earnings.

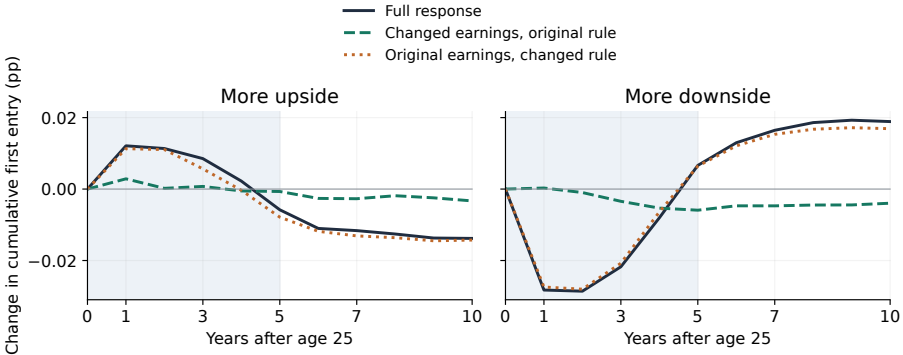
$$\text{More upside: } (U_y + d, L_y - d) \qquad \text{More downside: } (U_y - d, L_y + d).$$

- d transfers variance between the tails; it is not an income payment.
- **Years 1–5 (ages 26–30):** draw from the changed distribution each year. **Year 6 onward:** baseline. Households know this schedule.
- Parameters, starting states and opportunity/return draws remain fixed. The entry–risk coefficients are not targeted.

Follow 395 initial nonholders from age 25 for ten years. Shock size is not fitted to the entry coefficient.

▶ Cumulative first entry ▶ Saving at the same cash

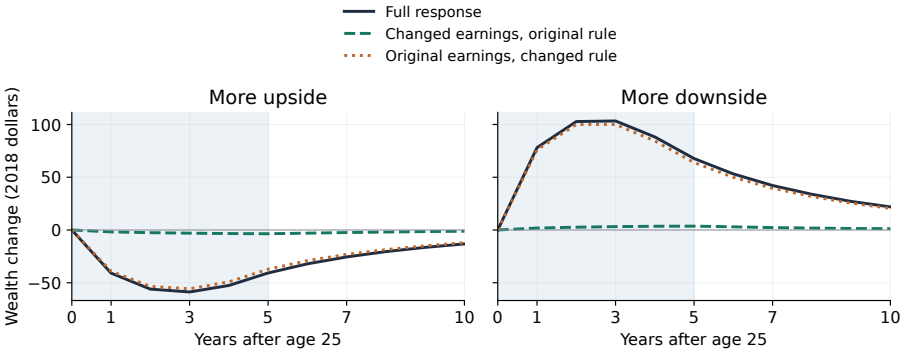
Cumulative first entry after the composition shock



Share who have entered at least once, shown as a change from baseline (pp).
Years 1–5: changed earnings distribution. Year 6 onward: baseline. Households know this schedule.

▶ Wealth ▶ Saving at the same cash

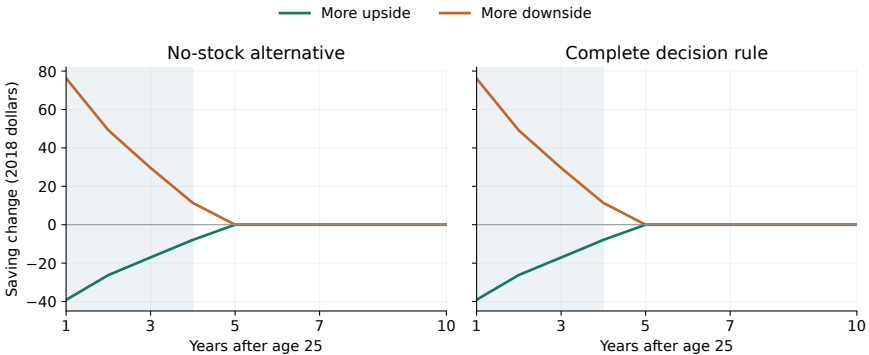
Lower saving reduces wealth available for later entry



At the same current cash, lower saving raises consumption and reduces future resources. Realized earnings change cash received. Saving determines how much is carried forward.

▶ Same-cash comparison ▶ Consumption and fees

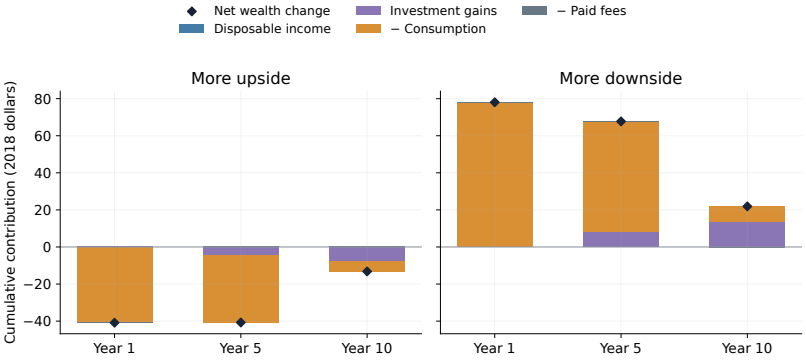
Saving falls at the same current resources



First-year saving falls by about $39, including when households choose not to hold stocks. Future earnings risk changes desired saving, even with expected earnings fixed.

▶ Where the money goes ◀ Back

Higher consumption accounts for the initial wealth decline



First-year response: consumption rises $40.81; fees rise $0.03.
Expected earnings stay fixed. Households spend more and leave less saving to finance later entry.

◀ Back ▶ One year or lasting prospects? ▶ Model assessment

Conclusion

- Earnings upside balance predicts stock-market entry conditional on mean growth, total variance and measured resources.
  - A one-SD increase in upside balance predicts 1.60 pp more annual entry.
- The association also appears for entry followed by subsequent ownership.
- The model reproduces the positive sign but understates the magnitude.
  - Realized earnings help finance entry.
  - Future earnings risk changes saving and later participation.

The entry–risk relationship provides a test beyond the financial outcomes used to estimate the model.

▶ Closest studies ▶ Skewness ▶ Appendix

Additional results and model details

Empirical evidence

- ▶ How are the measures constructed?
- ▶ What drives the differences across cells?
- ▶ Which cells are in the scatter?
- ▶ Why not regress on the two tails?
- ▶ What are the controls?
- ▶ Alternative earnings measures
- ▶ PSID evidence
- ▶ Which ownership sequences count?

Model mechanisms

- ▶ Why 58 cells?
- ▶ Earnings process equations
- ▶ The household optimization problem
- ▶ Two-cell numerical example
- ▶ Resources versus decision rules
- ▶ One year, five years or working life?
- ▶ All duration decompositions
- ▶ Financial fit and discrepancies

◀ Back

Earnings variation and transitions to zero wages

Across 58 cells: correlation with zero-wage share	Correlation
Total variance V	$+0.92$
Upside balance C	-0.85

Groups with more transitions to zero wages have higher variance and less upside balance.

- The wage measure includes transitions from positive annual wages to zero. Zero wages need not mean unemployment or dismissal.
- Annual earnings also reflect hours, job changes and wage changes. These records do not separate promotions from persistent losses.
- The CPS earnings data precede the participation outcomes. This cross-cell comparison does not identify a business-cycle shock.

◀ Back ▶ Does zero-wage risk absorb the result? ▶ Persistence and cycle evidence

Controlling for the frequency of zero wages

Same 27,419 initial nonholders, controls, weights and 30 industry–education clusters.

Coefficient, pp per SD	Baseline	Add zero-wage share
Upside balance C	1.601^{***}	2.420^{**}
	(0.506)	(0.911)
Zero-wage share	—	2.630
		(2.030)

The positive association remains after controlling for the frequency of zero wages.

The larger standard error reflects a demanding comparison among strongly related cell measures. The coefficient increase does not identify a causal channel.

Outcome: next-year stock market participation among initial nonholders. Clustered SEs in parentheses, t_{29} inference. $^{**}p < 0.05$, $^{***}p < 0.01$.

◀ Back ▶ Two cells at similar variance

Two cells with similar variance and different upside balance

Both age 25–44. Semivariances use age-adjusted annual arc wage growth.

Education	Trade Some college	Transport / utilities High school
Upper semivariance V^+	0.278	0.253
Lower semivariance V^-	0.383	0.409
Total variance V	0.661	0.661
Upside balance C	-0.106	-0.156
Zero annual wages next year	5.67%	7.75%
Linked earnings histories	2,493	802

Trade has less lower-tail and more upper-tail variation.

The difference in upside balance is about 1.5 SD. These descriptions do not identify the job events behind the differences.

◀ Back ▶ All 58 cells

Controlling for cyclical exposure

Add each measure of cyclical exposure separately to the baseline entry regression.

Additional control	Upside balance, pp	SE
None	1.601^{***}	0.506
Cell's mean wage growth in 2020	1.783^{***}	0.528
Mean-growth exposure to unemployment changes	1.612^{***}	0.479
Lower-semivariance exposure	1.701^{***}	0.533
Upside-balance exposure	1.662^{***}	0.487
All three unemployment exposures	2.104^{***}	0.452

The positive association remains after controlling for cyclical exposure.

PSID exposures: slopes of biennial cell moments on national unemployment changes, 1999–2019. Same 27,419 SIPP transitions, 30 clusters, CR1/t_{29}. $^{***}p < 0.01$.

◀ Back ▶ Permanent or temporary changes?

Persistent and temporary earnings changes

- **Annual CPS histories do not label the underlying event.** Wage changes can combine job moves, hours changes and changes in wage rates.
- **An exploratory comparison by the transitory share of earnings risk is sensitive to measurement.** Its positive interaction weakens when extreme earnings changes are trimmed.
- **Longer PSID histories discipline persistence in the model.** They do not establish which event caused a particular CPS cell's position in the scatter.

The evidence compares earnings distributions across worker groups.

It does not assign causal shares to promotions, permanent losses and temporary earnings changes.

◀ Back ▶ Model earnings components

Earnings cells and measured moments (1/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Agriculture	HS	25-44	0.305	0.508	12.2	623
Agriculture	HS	45-65	0.348	0.524	17.7	565
Construction	College	25-44	0.247	0.379	6.2	491
Construction	College	45-65	0.303	0.432	8.8	436
Construction	HS	25-44	0.299	0.468	10.4	1,950
Construction	HS	45-65	0.286	0.449	10.6	1,661
Construction	Some col.	25-44	0.282	0.421	7.7	867
Construction	Some col.	45-65	0.274	0.431	10.0	767
Manufacturing	College	25-44	0.162	0.233	2.9	1,520
Manufacturing	College	45-65	0.192	0.290	4.4	1,623

CPS baseline years 2013–2017. Survey-weighted moments; unweighted observation counts.

◀ Back ▶ Next cells

Earnings cells and measured moments (2/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Manufacturing	HS	25-44	0.201	0.326	5.1	1,936
Manufacturing	HS	45-65	0.201	0.324	6.2	2,694
Manufacturing	Some col.	25-44	0.201	0.296	3.9	1,228
Manufacturing	Some col.	45-65	0.170	0.269	4.3	1,478
Trade	College	25-44	0.234	0.361	5.4	1,822
Trade	College	45-65	0.302	0.420	7.3	1,627
Trade	HS	25-44	0.282	0.407	7.2	3,490
Trade	HS	45-65	0.269	0.410	8.1	3,328
Trade	Some col.	25-44	0.278	0.383	5.7	2,493
Trade	Some col.	45-65	0.242	0.348	6.0	1,855

CPS baseline years 2013–2017. Survey-weighted moments; unweighted observation counts.

◀ Back ▶ Next cells

Earnings cells and measured moments (3/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Transport/utilities	College	25-44	0.202	0.290	3.6	365
Transport/utilities	College	45-65	0.221	0.355	4.5	445
Transport/utilities	HS	25-44	0.253	0.409	7.7	802
Transport/utilities	HS	45-65	0.234	0.397	7.8	1,144
Transport/utilities	Some col.	25-44	0.280	0.447	11.5	635
Transport/utilities	Some col.	45-65	0.188	0.345	7.5	784
Information	College	25-44	0.170	0.262	3.2	473
Information	College	45-65	0.212	0.290	4.0	428
Information	HS	25-44	0.187	0.297	5.3	360
Information	HS	45-65	0.188	0.304	4.1	411

CPS baseline years 2013–2017. Survey-weighted moments; unweighted observation counts.

◀ Back ▶ Next cells

Earnings cells and measured moments (4/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Information	Some col.	25-44	0.176	0.220	2.1	357
Information	Some col.	45-65	0.212	0.319	5.9	368
Finance/real estate	College	25-44	0.189	0.291	3.7	1,799
Finance/real estate	College	45-65	0.300	0.416	6.6	1,546
Finance/real estate	HS	25-44	0.226	0.433	10.1	531
Finance/real estate	HS	45-65	0.236	0.355	6.8	683
Finance/real estate	Some col.	25-44	0.192	0.341	6.0	842
Finance/real estate	Some col.	45-65	0.229	0.371	7.5	923
Professional/business	College	25-44	0.201	0.300	4.2	3,676
Professional/business	College	45-65	0.280	0.404	7.6	2,986

CPS baseline years 2013–2017. Survey-weighted moments; unweighted observation counts.

◀ Back ▶ Next cells

Earnings cells and measured moments (5/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Professional/business	HS	25-44	0.298	0.486	10.6	1,602
Professional/business	HS	45-65	0.310	0.450	10.2	1,774
Professional/business	Some col.	25-44	0.298	0.437	7.4	1,663
Professional/business	Some col.	45-65	0.242	0.389	8.1	1,683
Education/health	College	25-44	0.191	0.271	3.5	6,966
Education/health	College	45-65	0.192	0.304	4.8	6,270
Education/health	HS	25-44	0.297	0.430	8.1	1,581
Education/health	HS	45-65	0.212	0.335	6.9	2,458
Education/health	Some col.	25-44	0.237	0.335	5.1	2,989
Education/health	Some col.	45-65	0.205	0.333	5.8	3,038

CPS baseline years 2013–2017. Survey-weighted moments; unweighted observation counts.

◀ Back ▶ Next cells

Earnings cells and measured moments (6/6)

$V = V^+ + V^-$; $C = V^+ - V^-$. HS: high school or less. Zero: no wages next year.

Industry	Education	Age	V^+	V^-	Zero %	Histories
Leisure/hospitality	HS	25-44	0.374	0.546	16.0	616
Leisure/hospitality	HS	45-65	0.386	0.560	14.8	688
Leisure/hospitality	Some col.	25-44	0.370	0.552	13.7	389
Public administration	College	25-44	0.121	0.150	1.0	1,339
Public administration	College	45-65	0.159	0.233	2.9	1,427
Public administration	HS	45-65	0.176	0.302	5.5	653
Public administration	Some col.	25-44	0.178	0.235	3.0	769
Public administration	Some col.	45-65	0.173	0.283	5.0	945

CPS baseline years 2013–2017. Survey-weighted moments; unweighted observation counts.

◀ Back ▶ First cells

A direct regression on upper and lower semivariances

Among initial nonparticipants, regress next-year participation on

$$\alpha + b_+ V_c^+ + b_- V_c^- + \text{mean growth, resources, demographics and fixed effects.}$$

This linear specification omits $\sqrt{V}$. All other controls and the sample are unchanged.

Direct coefficient	Estimate	Clustered SE
Upper semivariance V^+	$+0.331^*$	(0.167)
Lower semivariance V^-	-0.324^{**}	(0.154)

Units: percentage points of participation per 0.01 growth-squared unit of the indicated semivariance, holding the other fixed.

More upper variation predicts more participation; more lower variation predicts less.

One linear regression, equivalently parameterized by C, V. These are not local derivatives of the nonlinear baseline. $N = 27,419$; 30 industry–education clusters; t_{29} tests. $^*p < 0.10$, $^{**}p < 0.05$.

◀ Back ▶ Controls and sample

Contribution

How earnings vary helps explain who becomes an investor.

- Upside balance predicts entry at similar mean growth, variance and measured resources.
- The measure describes earnings asymmetry, without using stock returns. It differs from exposure to joint downturns in earnings and stocks.
- The entry relationship tests portfolio models beyond the financial outcomes used for estimation.

▶ Measurement comparisons ▶ Closest studies ▶ Related work in progress

Literature: resources and stock-market entry

- **Resources and ownership transitions.**
  Briggs et al. (JFE, 2021); Bonaparte et al. (Management Science, 2025)
  Lottery windfalls raise entry among nonholders. Income and wealth changes also help explain entry and exit.

- **This paper measures earnings opportunities before entry.**
  Earlier earnings distributions, measured in a separate survey, predict entry and entry followed by later ownership.

- **The model tests the same entry relationship.**
  Entry–risk coefficients are excluded from fitting. The model separates changes in resources along earnings paths from changes in saving and investment decisions.

▶ Closest empirical studies ▶ Model comparison

Organization of the analysis

1. **Measure:** where does earnings variation come from?
2. **Evidence:** who starts investing, and who keeps holding?
3. **Checks:** earnings measurement, sorting and reporting
4. **Model:** how do earnings paths and saving decisions shape entry?

Specification and measurement checks

Each check changes the baseline separately. The first block requires positive financial assets and valid risky shares at both dates.

Concern	Specification	Estimate, pp (SE)	Nonholders
	Baseline: valid shares at both dates	1.646^{**} (0.599)	24,235
Group-year conditions	Industry $\times$ year; education $\times$ year	1.663^{**} (0.619)	24,235
Persistent group sorting	Instead: industry $\times$ education effects	1.865^{**} (0.875)	24,235
Reporting and household changes	Reported, self-reported, stable households	2.557^{***} (0.555)	11,229
Realized earnings: a separate common sample of 23,261 nonholders			
	Baseline	1.518^{**} (0.587)	23,261
	Add annual earnings level and change	1.574^{**} (0.586)	23,261

Both broad and nonretirement denominators must be positive; their risky shares must lie in $[0, 1]$.

▶ Sorting ▶ Samples

Evidence from the PSID

The Panel Study of Income Dynamics (PSID) provides evidence from another survey and period: biennial entry among initial nonholders; baseline years 1999–2017.

Age range	Upside balance (pp)	SE
Full sample: 25–65	-0.18	0.39
Exploratory age split		
25–44	2.42^{**}	1.11
45–65	-0.98	1.01

No clear full-sample association; a positive estimate among younger workers.

PSID uses different households. Some outcomes occur before the CPS earnings data, so this is not the same timing design as SIPP.

▶ Later-period age comparison ▶ Period differences

Earnings upside balance and participation decisions

- **Main result:** more upside-oriented variation predicts more next-year entry at similar variance and resources.
- **Later ownership:** the association also appears for entry followed by ownership at later annual observations.
- **Economic question:** how does the earnings environment relate to a nonholder's subsequent investment decision?

The model tests a familiar resource mechanism against this entry pattern.

▶ Entry and retention

Same variance, different chances of reaching the threshold

Earnings innovation	Symmetric	More upside
Possible loss	$-\$1{,}000$ (50%)	$-\$500$ (80%)
Possible gain	$+\$1{,}000$ (50%)	$+\$2{,}000$ (20%)
Mean / standard deviation	\$0 / \$1,000	\$0 / \$1,000
Entry if \$1,500 short	0%	20%
Entry if \$500 short	50%	20%

The gain matters when it carries a household across the threshold.

The sign depends on where households start. Cells are not needed for this mechanism.

Illustrative dollar innovations around common expected earnings, not estimated responses.

◀ Back ▶ Derivation

Model estimation and the untargeted entry test

1. Estimate earnings processes from CPS and PSID earnings moments.
2. Fit five financial parameters to eight SIPP financial outcomes.
3. Solve the same saving and participation problem under cell and pooled earnings dynamics.
4. Apply the empirical entry regression to those predictions.

Untargeted test: do measured earnings environments generate the observed upside balance–entry relationship?

Earnings-risk coefficients never enter the fit. Related comparison: Gálvez & Paz-Pardo (2026).

▶ Initial states ▶ Published precedents

Four adjustments connect each cell to its earnings data

Cell adjustment	What it changes
Departure risk	Probability of leaving wage work
Growth	The predictable earnings path with age
Shock scale	The size of transitory variation and reporting error
Asymmetry tilt a_c	Relative weights on favorable and adverse persistent and temporary shocks

Fit all four adjustments jointly to four CPS earnings moments:

zero-wage frequency, mean growth, upper partial variance and lower partial variance.

PSID longer-horizon moments discipline the shared career process. Stock-market entry coefficients never enter earnings estimation.

▶ How the tilt works ▶ Exact cell mapping

Asymmetry in persistent and temporary shocks

Persistent earnings combine ordinary innovations with occasional jumps.

$$\eta_{t+1} = \rho\eta_t + \zeta_{t+1} + \underbrace{J^+ I^+_{t+1}}_{\text{promotion}} - \underbrace{J^- I^-_{t+1}}_{\text{earnings loss}} - m.$$

ρ: persistence; ζ: small normal innovation; m: centering drift.

J^+, J^-: jump sizes; I^+, I^-: indicators for promotion or loss.

Promotion and loss are mutually exclusive. Their sizes and probabilities can differ.

A positive cell tilt a_c shifts relative weight toward favorable shocks.

$$\text{Promotion weight } \propto e^{a_c}, \qquad \text{Loss weight } \propto e^{-a_c}.$$

Temporary pay shocks follow a mixture distribution. The same tilt shifts their relative weights toward the upper tail.

▶ Exact persistent tilt ▶ Temporary-shock tilt

Current resources and future earnings expectations

Realized resources	Earnings already received add to cash available for consumption, saving and participation fees.
Expectations	The future earnings distribution changes how valuable a financial buffer and stock investment are today.
Saving over time	Today's saving becomes invested wealth. Its returns and next year's income become next year's cash.

A decision rule gives the optimal choice for each current state.

At identical cash and other states, different future risks can change consumption, saving and entry.

Expected pay can stay fixed. The value of a buffer also depends on what happens in bad earnings outcomes.

▶ Continuation values ▶ Earnings and decision rules

Two-period illustration: same mean and variance

Earnings today and next year follow one of these two processes.

All amounts are in thousands of dollars.

	Upside cell (U)	Downside cell (D)
Ordinary outcome	85 with 80% chance	115 with 80% chance
Tail outcome	160 with 20% chance	40 with 20% chance
Mean / standard deviation	100 / 30	100 / 30

U has a rare large gain; D has a rare large loss.

Same log utility and patience; entry costs 1.5. Safe saving earns 0%; stocks return -50% or $+70\%$, each with 50% chance, independently of earnings.

These are illustrative cells, separate from the fitted 58-cell model.

▶ Full assumptions ▶ Same cash, different prospects

Two-period illustration: expectations at the same cash

Give both households 100 in cash today. Only the future distribution differs.

Future earnings process	Consume today	Save safely	Enter stocks?
Upside cell (U)	96.96	3.04	No
Downside cell (D)	92.77	7.23	No

The downside household saves 4.19 more as a buffer.

Both expect 100 tomorrow. D can earn only 40, while U earns at least 85. The buffer is more valuable in D despite equal means and variances.

This isolates the response to future risk. Current cash, preferences, entry fees and investment opportunities are identical.

All amounts are in thousands of dollars. Optimal choices under illustrative parameters.

◀ Back ▶ Let earnings arrive ▶ Full assumptions

Two-period illustration: realized earnings and participation

Start with no inherited assets. Current and future earnings are independent draws from the same cell process. Amounts are in thousands of dollars.

Current outcome	Chance	Cash	Optimal saving	Enter?
U: ordinary	80%	85	0	No
U: large gain	20%	160	32.22 in stocks	Yes
D: large loss	20%	40	0	No
D: ordinary	80%	115	14.02 safely	No

The large gain finances stock entry despite the fee.

U: 20% enter; mean saving is 6.44. D: 0% enter; mean saving is 11.22.

The saving gap reflects differences in both realized earnings and expectations. It is not an isolated precautionary-saving effect.

◀ Back ▶ Full assumptions

Apply the same regression to model entry probabilities

	Data	Model
Outcome	Entry observed: 0 or 1	Predicted entry probability
Households	The same 17,606 initial nonholders	
Right-hand side	Same risk measures, controls, fixed effects and weights	

Average over earnings, opportunities and costs, conditioning each observation on next-year wage work and positive assets.

Compare predictions under cell earnings processes with the data.

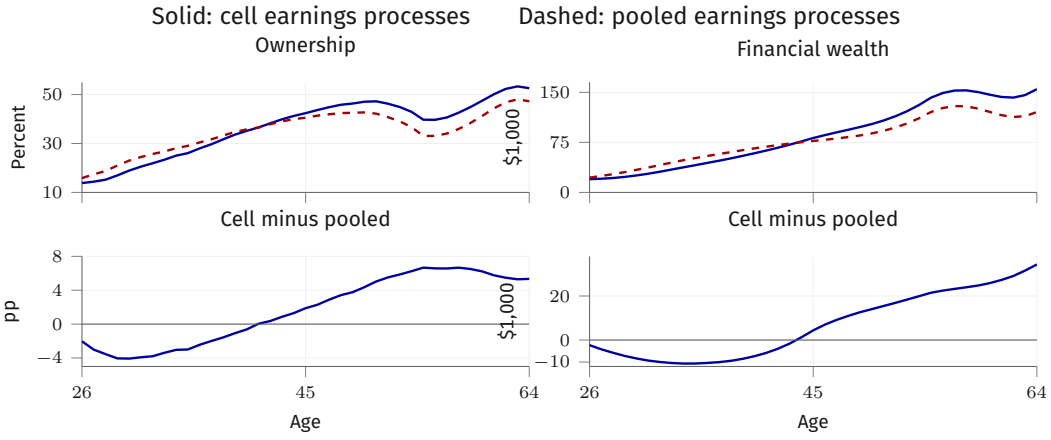
Subtract pooled predictions to describe the contribution of cell earnings processes.

The matched data coefficient is 1.816 pp; the full-sample headline is 1.601 pp.

▶ Why probabilities do not shrink coefficients ▶ Matched sample

Ownership and wealth paths

Simulated paths for the same 135 age-25 observations in two college groups



One-year, five-year and working-life changes

Five years is an illustrative horizon, not an estimated duration.

Duration	Affected ages	What follows
One year	26	Baseline earnings from age 27
Five years	26–30	Baseline earnings from age 31
Remaining working life	26–64	Retirement income unchanged

The same annual shock at each age affected in both scenarios; expected disposable earnings and dollar variance stay fixed.

Same 395 initial nonholders and fixed parameters; risk coefficients remain untargeted.

Households know the schedule. Starting portfolios are given; the first new choice follows year 1 earnings.

◀ Back ▶ Entry comparison ▶ Wealth comparison

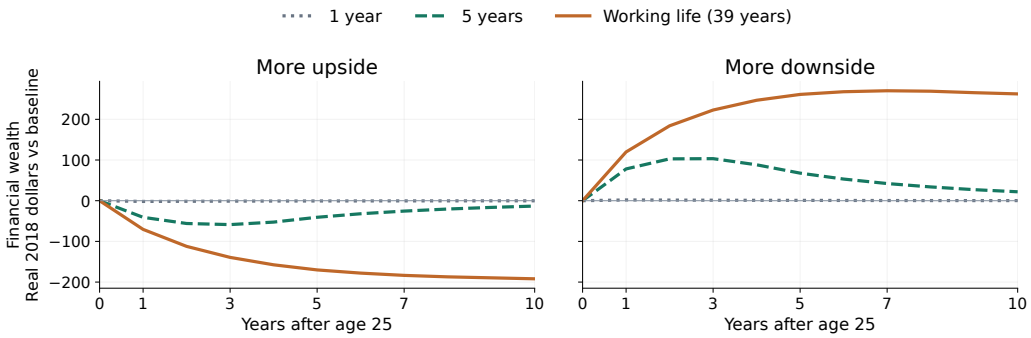
Entry responses by duration



The five-year reversal depends on how long earnings prospects change.
One-year entry is near zero and grid-sensitive. Cumulative entry under lasting upside fades toward zero. Its later annual first-entry differences are negative.

◀ Back ▶ Wealth ▶ Annual first entry

Wealth responses by duration



Lasting prospects sustain the saving and wealth response.
At year 10, lasting upside lowers wealth by about $192; lasting downside raises it by $262. The five-year gaps are about $13 and $22.

◀ Back ▶ Why wealth and entry differ ▶ All comparisons

Wealth accumulation and cumulative first entry

Illustrative upside shift: expected earnings stay fixed.

- **At the same cash and state,** different future risks make entry slightly more attractive and reduce the desired savings buffer.
- **Saving then changes future resources.** More consumption leaves less invested wealth and lower investment gains.
- **Cumulative entry retains the early entrants.** Fewer first-time entrants later roughly offset the initial extra entrants.

Five years: the incentive to enter ends, but the smaller savings buffer remains. **Lasting change:** both forces continue.

◀ Back ▶ Entry at the same cash ▶ Annual entry ▶ All diagnostics

Model results and remaining discrepancies

- Earnings gains can give households the resources needed to enter.
- Differences in earnings across cells strengthen the untargeted entry association, mainly through realized resources.
- Future earnings risk changes saving at the same current cash and therefore the resources available for later entry.
- Upside-balance coefficient: 0.241 pp with cell processes, against 1.816 pp in the matched data.

Nonholders accumulate too much wealth and former holders re-enter too slowly.

▶ Cell versus pooled ▶ Financial discrepancies ▶ Saving illustration

Earlier evidence on income risk and portfolios

Study	Empirical risk measure
Guiso, Jappelli & Terlizzese (1996, AER)	Subjective income uncertainty
Angerer & Lam (2009, JF)	Permanent and transitory income variability
Fagereng, Guiso & Pistaferri (2018, ReStud)	Firm-transmitted uninsurable wage risk

This paper studies the balance of above- and below-mean earnings variation and subsequent entry.

Different income-risk measures and portfolio outcomes imply different empirical comparisons.

◀ Back ▶ Theory and measurement

Removing age profiles and centering are different steps

Group G: industry $\times$ education. Cell c: group $G \times$ age band.

1. Measure annual wage growth, retaining zero wages next year.

$$g_{i,t+1} = \frac{2(y_{i,t+1} - y_{it})}{y_{i,t+1} + y_{it}}, \qquad y_{it} > 0, \quad y_{i,t+1} \geq 0.$$

y is annual nominal wages. When next-year wages are zero, $g = -2$.

2. Remove the age profile within the broader group G.

$$\epsilon_{i,t+1} = g_{i,t+1} - \widehat{m}_G(a_{it}), \qquad \widehat{m}_G(a) \text{ is a weighted cubic in age.}$$

The regression includes an intercept, so residuals average to zero within G.

3. Center within the finer age-band cell c.

$$\mu_c = \mathbb{E}_c[\epsilon], \qquad \text{deviation} = \epsilon - \mu_c.$$

A zero mean across all ages in G does not imply zero means in each age band. Semivariances use these within-cell deviations.

◀ Back ▶ Growth measure ▶ SIPP regression controls

Retirement contributions

Complementary evidence on participation in another saving vehicle. These contribution outcomes are not calibration targets.

Initial non-IRA stock ownership	Upside balance (pp)	SE
Nonholders	0.22	0.33
Holders	6.36***	2.19
Holder minus nonholder	6.14**	2.33

Outcome: any private IRA or annuity contribution, not the amount saved or the account's equity share.

Complementary evidence on household purchases

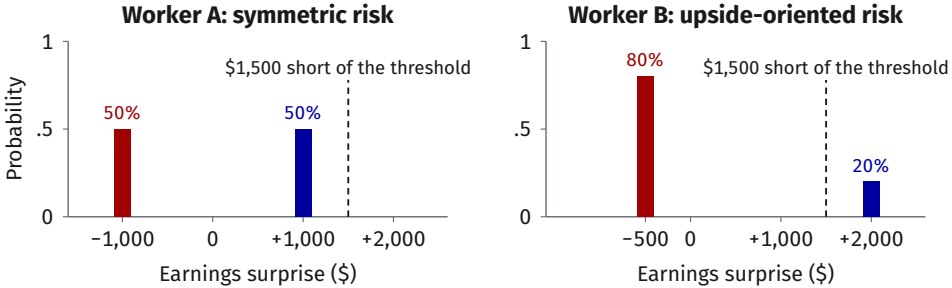
Complementary evidence on occasional purchases. Not used to fit the model.

Asinh dollar scale	Furnishings/ equipment $1,000	Selected nondurables/ services $10,000	Food $10,000
Upside balance, holding variance fixed			
Initial nonholders	0.007	0.001	0.002
	(0.013)	(0.008)	(0.005)
Initial holders	0.114***	0.012	0.009
	(0.041)	(0.020)	(0.017)
Holder minus nonholder	0.107**	0.011	0.006
	(0.046)	(0.024)	(0.019)
Average using baseline group shares	0.027**	0.003	0.004
	(0.012)	(0.006)	(0.004)
Variance, holding upside balance fixed			
Initial holders	0.021***	0.003	0.001
	(0.006)	(0.004)	(0.003)
33,353 observations; 30 industry–education clusters			

Furnishings: positive among holders. Other baskets: imprecise.

▶ Why study consumption? ▶ Inference

Earnings gains near the threshold



Entry probability: **0%** for A **20%** for B

Life-cycle model

Earnings risk and stock market participation

Saving, entry opportunities and holding costs

The household must finance consumption, saving and any fee

M_t: current cash; c_t: consumption; A_t: saving carried forward.

$$c_t + A_t + D_t K_t = M_t, \qquad K_t = \kappa_R \xi_t.$$

D_t: stockholding indicator; K_t: fee offer; κ_R: mean fee; ξ_t: mean-one multiplier.

$$M_{t+1} = A_t\{R_{f,t+1} + \alpha_t(R^s_{t+1} - R_{f,t+1})\} + Y_{t+1}.$$

α_t: stock share; R_f, R^s: gross safe and stock returns; Y: disposable income.

- Positive consumption, no borrowing: $c_t > 0$, $A_t \geq 0$.
- Access $a_t \in \{0, 1\}$ is known before choice: $D_t \in \{0, 1\}$ and $D_t \leq a_t$.
- No short positions: $0 \leq \alpha_t \leq D_t$. Funded stocks: $\alpha_t A_t \geq \$1 \times D_t$.

◀ Back ▶ Access and fees ▶ Optimization

An investment opportunity permits entry; a fee makes it costly

d_t: prior stockholding; a_t: opportunity to invest now.

$$\Pr(a_t = 1 \mid d_t) = d_t + (1 - d_t)q.$$

- A previous nonholder receives a fresh opportunity with probability q.
- A previous holder can choose again. Exit returns the household to the nonholder opportunity process.
- An opportunity permits entry; it does not require participation.

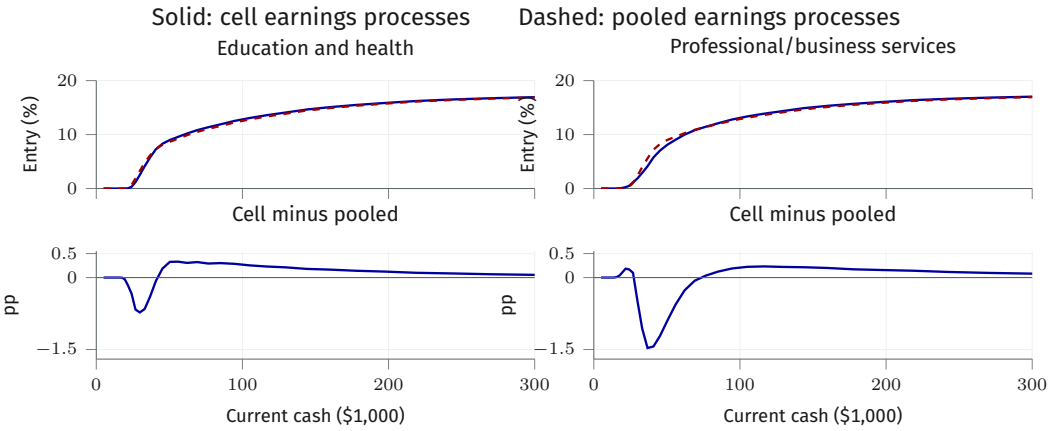
$$K_t = \kappa_R \xi_t, \qquad \mathbb{E}[\xi_t] = 1.$$

K_t: fee offer; κ_R: mean offer; ξ_t: independent lognormal multiplier.

The household pays the fee only if it chooses a funded stock position.

▶ Structural parameter estimates ▶ Offers versus payments ▶ Permanent access

Entry probabilities by current cash



Targeted financial moments

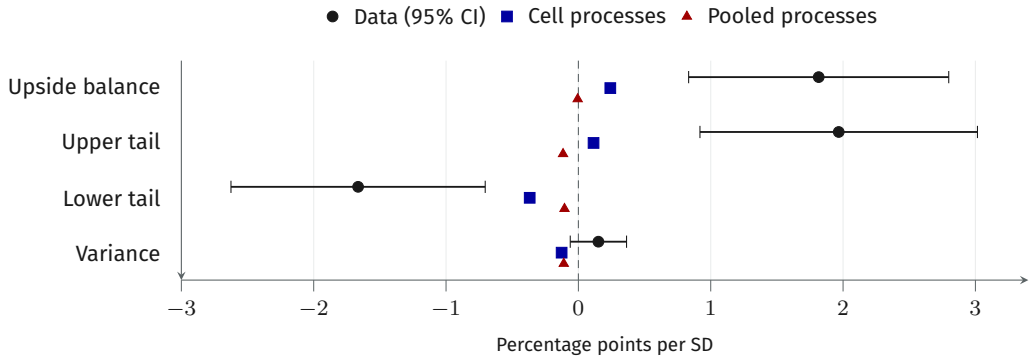
Moment	Unit	Data	SE	Model
Entry: low initial wealth	%	2.93	(0.26)	3.18
Entry: middle initial wealth	%	5.84	(0.36)	5.51
Entry: high initial wealth	%	10.36	(0.48)	10.44
Retention: low initial equity share	%	66.19	(1.43)	67.75
Retention: high initial equity share	%	76.31	(1.23)	74.34
Next equity share: all initial holders	%	33.38	(0.84)	34.40
Wealth change: initial nonholders	$	3,939	(778)	9,941
Wealth change: initial holders	$	1,946	(5,177)	511

Main discrepancy: excess nonholder wealth growth

◀ Back ▶ Additional diagnostics ▶ Alternative fit

Entry estimates from the model

Data and model use the same 17,606 initial nonholders; coefficients are untargeted



All rows use the CPS SD of upside balance. Upper- and lower-tail estimates are local derivatives of the same regression.

◀ Back ▶ Permanent alternative

Earnings distributions in portfolio models

Study	Existing mechanism or comparison
Cocco, Gomes & Maenhout (2005)	Labor income, borrowing constraints and downside earnings risk
Catherine (2022)	Earnings downside when stock returns are poor
Shen (2024)	Skewed earnings risk, portfolio choice and participation costs
Gálvez & Paz-Pardo (2026)	Rich earnings dynamics and risk regressions in both data and model

This paper tests whether the fitted model reproduces the observed entry–risk relationship.

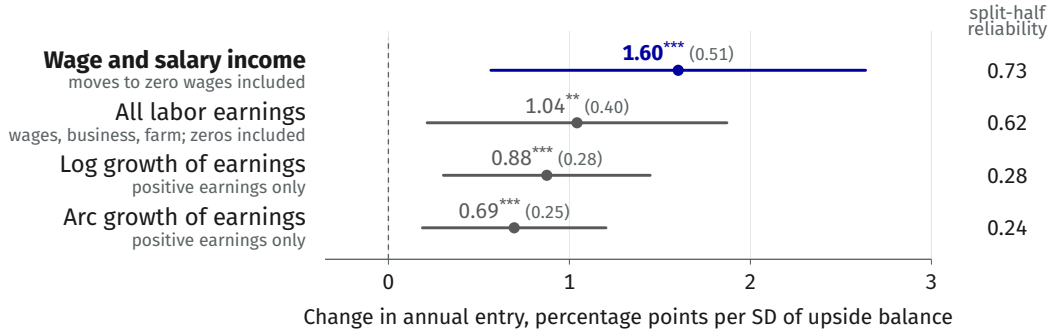
◀ Back ▶ Detailed comparison

Upside balance at fixed variance

	Symmetric risk	Upside-oriented risk
Lower outcome	$-1{,}000$ (50%)	-500 (80%)
Upper outcome	$+1{,}000$ (50%)	$+2{,}000$ (20%)
Lower partial variance	500,000	200,000
Upper partial variance	500,000	800,000
Total variance	1,000,000	1,000,000
Upside balance	**0**	**600,000**

Upside balance: upper minus lower partial variance.

Entry estimates across earnings measures



Positive entry estimates across all four measures. The wage measure that includes zeros has the highest reliability.

▶ Zeros and growth ▶ Skewness

Observed initial states

21,188 SIPP worker-year pairs, initial ages 25–63

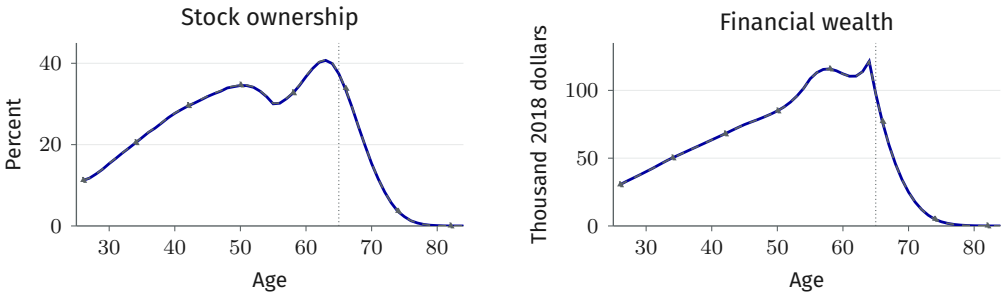
Input	Use in the model
Wages, wealth, ownership and shares	Latent earnings from wages; observed financial positions
Taxes and own pension contributions	Wages net of taxes and contributions
Own retirement income	PSID Social Security and pensions by education and age
Calendar returns	Returns on inherited portfolios in observed transitions

Future return expectations: stationary.

Conditional life-cycle profiles

Conditional projections for 452 age-25 observations in 29 groups

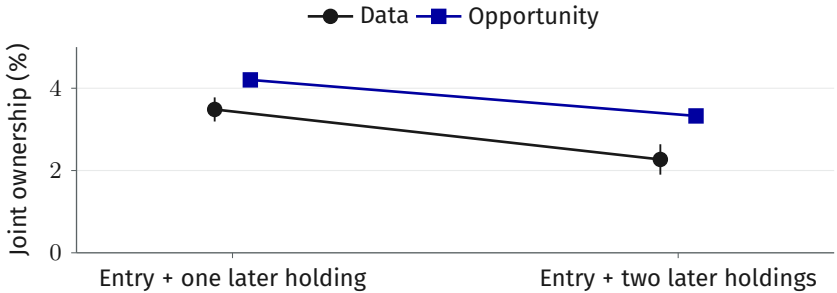
Solid: 141-node grid Dashed triangles: 201-node grid



Numerical accuracy: almost identical paths on finer cash and return grids.

Entry and subsequent ownership

Untargeted predictions for the same follow-up samples in the data and model



Sustained ownership levels remain overpredicted.

▶ Common-cohort estimates

Samples

Sample	Observations
CPS linked earnings histories	88,862
CPS histories with zero wages the next year	5,490
SIPP full entry sample	27,419
SIPP valid-share sample at both dates	31,569
Worker-year pairs used to estimate the model	21,188
Entry sample shared by the data and model	17,606

The 31,569 allocation observations require positive broad and nonretirement financial assets and valid risky shares at both dates: 24,235 nonholders and 7,334 holders.

Entry is measured for workers; assets are measured for households. Mixed-asset mutual funds included; retirement-account allocation is unobserved.

◀ Back ▶ Appendix index

Arc wage growth

$$g_{i,t+1} = \frac{2(y_{i,t+1} - y_{it})}{y_{i,t+1} + y_{it}}.$$

- Positive wages initially; zero wages the next year included: $g = -2$
- Remove a weighted cubic age profile within industry–education groups
- Then center within the finer industry–education–age-band cells
- Zero wages: nonemployment or a different income source, not necessarily dismissal

◀ Back ▶ Appendix index

Relation to skewness

	Upside balance C	Conventional skewness
Construction	Signed squared deviations	Cubed deviations, divided by SD^3
Interpretation	Upper minus lower contributions to variance	Standardized third moment
Extreme observations	Squared weighting	Cubed weighting

Both describe asymmetry. They are closely related in these data.

Why this measure? Additive variance decomposition; higher measured reliability (0.73 vs. 0.39).

These results do not show that upside balance predicts entry better than skewness.

◀ Back ▶ Skewness comparisons

Upside balance and skewness: the same controls

Zero-inclusive wage measure	Upside balance	Skewness
Each measure separately	1.601^{***}	0.912^{**}
	(0.506)	(0.414)
Both measures together	2.585	-0.726
	(1.670)	(1.195)
Split-half reliability	0.73	0.39

Same 27,419 observations; all headline controls, including V and $\sqrt{V}$.

Partial correlation given $V, \sqrt{V}$: 0.91. The joint regression does not precisely distinguish the two associations.

Estimates are in pp per SD of each measure. Clustered SEs; t_{29} tests.

◀ Back ▶ Auxiliary total-earnings comparison

Alternative earnings measures

Construction	Upside balance (pp)	SE
Wages, zeros included	1.601^{***}	0.506
Total labor earnings, zeros included	1.042^{**}	0.405
Real log growth, positive earnings	0.875^{***}	0.279
Nominal arc growth, positive earnings	0.695^{***}	0.248

Controls: $V, \sqrt{V}, \mu, \mu^2$, household characteristics
Industry, education, age-band and year fixed effects

◀ Back ▶ Appendix index

Statistical inference

- **Coefficient tests:** 30 industry–education clusters, CR1/t_{29}
- **Wild-cluster bootstrap:** $p = 0.039$
- **Earnings-measurement uncertainty:** re-estimate CPS moments in separate checks
- Restricted-sample tests use their actual cluster counts. Stars use unadjusted coefficient tests.

◀ Back ▶ Appendix index

Significant variance terms can give a small local slope

Holding upside balance and controls fixed, the variance terms in unstandardized units are

$$f(V) = b_V V + b_S \sqrt{V}, \qquad b_V < 0, \quad b_S > 0.$$

$$f'(V) = b_V + \frac{b_S}{2\sqrt{V}}.$$

A concave relationship: the fitted entry association first rises with variance, flattens, then falls. The two coefficients describe one curve.

Earnings volatility	Lower quartile	Mean	Upper quartile
Local response (pp)	$+0.438^{***}$	$+0.104$	-0.212
	(0.135)	(0.116)	(0.136)

Notes: The regression includes C. Responses use the SD of upside balance; SEs account for coefficient covariance. A small slope at the mean can coexist with significant variance terms.

◀ Back ▶ Appendix index

Person fixed effects and within-person variation

Historical risk measures are fixed within each earnings cell.

Person FE would use changes in assigned risk when a worker changes cell; full cell FE would absorb the risk measures.

Repeated-observation comparison	Pooled	Within person
Upside balance (pp per SD)	1.167	0.893
Clustered SE	(0.688)	(0.933)
Initial nonholder observations	14,055	14,055

The within-person estimate is imprecise ($p = 0.347$).

5,717 people; 1,523 change cell. Sorting remains a possible source of the headline association.

The comparison demeans observations within person, then applies survey-weighted regression, with 30 industry–education clusters. It is not an exact weighted person-FE estimate.

◀ Back ▶ Other sorting checks ▶ Index

Entry and subsequent ownership

Upside balance estimate (pp)	One follow-up	Two follow-ups
Any entry	2.09***	3.31**
Entry and later holdings	0.98*	1.89**
Entry with an interruption	1.11**	1.42
Initial nonholders	9,236	3,254

The two joint outcomes divide entrants into subsequent holders and those with an interruption. Retention estimates among entrants are imprecise. Follow-up selection and weaker three-date reporting checks remain concerns.

◀ Back ▶ Appendix index

Period differences and age groups

Young-worker SIPP entry gap (pp)	Estimate
Original period gap	2.885
After symmetric overlap weighting	2.623
Change in gap	-0.262
SE of change	0.294
p-value of change	0.381

PSID: all ages, 1999–2017 baselines: -0.18 (SE 0.39).
Ages 45–65, 2009–2017 baselines: 0.44 (SE 1.30). All-age evidence is inconclusive.

◀ Back ▶ Appendix index

Exit among initial holders

This sample consists of people who already own stocks.

Next-year exit among initial holders	Upside balance estimate
Exit coefficient	$+3.72$ pp
Clustered standard error	1.71 pp
Wild-cluster bootstrap p-value	0.12

The estimate points toward more exit, but inference is sensitive.

More marginal investors entering and later exiting is one possible explanation. This regression does not test that channel.

▶ Holder outcomes ▶ Selection and interpretation

Initial-holder outcomes

Initial-holder outcome	Upside balance (pp)	SE
Next-period ownership / retention	-3.72^{**}	1.71
Next risky share, controlling initial share	-0.51	0.94

- Retention wild-cluster bootstrap $p = 0.12$.
- All initial holders, including subsequent exits
- Initial-holder retention differs from survival among new entrants

◀ Back ▶ Appendix index

Participation and household spending

Purpose of the expenditure evidence: compare the portfolio pattern with occasional purchases and regular spending. It is neither an entry robustness test nor a calibration target.

A simple benchmark: two dates, log utility, cash m, entry fee κ, saving and participation $d \in \{0, 1\}$.

$$c^*(m, d) = \frac{m - \kappa d}{1 + \beta}, \qquad \Delta \mathbb{E}[c] = -\frac{\kappa}{1 + \beta} \Delta \Pr(d = 1)$$

at fixed mean resources. More households can enter with only a small change in average consumption.

Empirical interpretation. Furnishings are occasional purchases; their positive primary estimate among holders does not establish a difference from other baskets. Dollar and original-construction estimates are imprecise.

The life-cycle model has one consumption good. It does not estimate separate durable-purchase decisions or total saving from IRA participation.

◀ Back ▶ Retirement contributions ▶ Index

The participation threshold follows from optimal saving

m: cash; c: consumption; $d \in \{0, 1\}$: entry; κ: fixed fee; β: discount factor.

With log utility, let $b > 0$ be the optimized gain in expected log portfolio return from stock access.

$$V_d(m) = \max_{0<c<m-\kappa d} \{\log c + \beta[\log(m - \kappa d - c) + bd]\}.$$

$$c_d(m) = \frac{m - \kappa d}{1 + \beta}, \qquad V_1 - V_0 = (1 + \beta) \log\left(1 - \frac{\kappa}{m}\right) + \beta b.$$

$$d^*(m) = 1 \iff m \geq m^* = \frac{\kappa}{1 - \exp[-\beta b/(1 + \beta)]}.$$

The fee absorbs a smaller share of cash when resources are higher.

This illustration has no terminal labor income. The life-cycle model adds future earnings uncertainty and repeated choices.

◀ Back ▶ Stationary selection formulas

A separate fixed-wealth illustration: who enters and stays?

Hold wealth w fixed. Draw earnings Y independently each date from distribution F.
κ: participation fee; h: safe terminal income; $b > 0$: optimized log-return gain.

$$w^* = \frac{\kappa}{1 - e^{-b}} - h, \qquad \pi(w, F) = \Pr(Y \geq w^* - w).$$

w^* is the required resource threshold; π is the participation probability at a given w, F.

$$\text{Entry} = \frac{\mathbb{E}[\pi(1-\pi)]}{\mathbb{E}[1-\pi]}, \qquad \text{Retention} = \frac{\mathbb{E}[\pi^2]}{\mathbb{E}[\pi]}.$$

$$\text{Entry + } k \text{ later holdings} = \frac{\mathbb{E}[(1-\pi)\pi^{k+1}]}{\mathbb{E}[1-\pi]}.$$

Expectations average across households with different w, F; k counts later holding dates.

These formulas isolate selection when wealth does not evolve.
The previous two-date model optimizes saving. This static illustration does not.

◀ Back ▶ Two-date mechanism ▶ Index

Cell definition and sample support

Industry, education and age distinguish work environments, skills and career stages. Broad cells retain enough earnings histories to measure the tails.

The 30 industry–education groups have two age bands: 60 possible group–age combinations. 58 have empirical support; two use the other age band in the model.

Sample size with narrower age groups	Current 58	Candidate 580
Median observations per cell	1,284	111
Median effective sample size	787	72
Cells meeting the sample-size requirement	58	50

Split each cell into ten narrower age groups. About 91% fail the sample-size requirement.

Finer cells need more data or estimates that share information across groups.

◀ Back ▶ Model mapping

Full entry projections: upside balance and local tail contrasts

Upper- and lower-tail estimates are local derivatives of the same regression.

Entry estimate (pp)	Data	SE	Cell	Pooled	Net
Upside balance	1.816^{***}	(0.480)	0.241	-0.005	0.247
Upper tail	1.967^{***}	(0.513)	0.115	-0.116	0.230
Lower tail	-1.665^{***}	(0.470)	-0.368	-0.105	-0.263
Variance	0.151	(0.104)	-0.127	-0.110	-0.016

- Pooled: both tail estimates negative. Cell: upper positive, lower negative.
- Neither relationship is targeted. Both cell estimates remain much smaller than in the data.

▶ Process and policy contributions ▶ Coefficient figure ▶ Why the test is informative

Selected references

- Catherine (2022). Countercyclical Labor Income Risk and Portfolio Choices over the Life Cycle. *Review of Financial Studies*.
- Catherine, Sodini & Zhang (2024). Countercyclical Income Risk and Portfolio Choices: Evidence from Sweden. *Journal of Finance*.
- Shen (2024). Countercyclical Risks, Consumption, and Portfolio Choice: Theory and Evidence. *Management Science*.
- Bonaparte, Korniotis, Kumar, Michaelides & Zhang (2025). Stock Market Ownership Transitions. *Management Science*.
- Gálvez & Paz-Pardo (2026). Richer earnings dynamics, consumption and portfolio choice over the life cycle. *Journal of Financial Economics*.
- Briggs, Cesarini, Lindqvist & Östling (2021). Windfall gains and stock market participation. *Journal of Financial Economics*.

Closest studies and this paper's contribution

Study	Empirical object
Catherine, Sodini & Zhang (2024)	Current risky shares including zeros, with participation decomposition. Unconditional and cyclical earnings skewness.
Shen (2024)	Changes in risky shares in the stockholder sample, related jointly to skewness and variance.
Bonaparte et al. (2025)	Actual ownership transitions related to income and wealth changes.
Gálvez (2023, working paper)	Participation responses following persistent earnings shocks.

Here: earlier CPS earnings upside balance and later SIPP entry among initial nonholders, at similar variance and resources.

Entry can include former owners returning to the market.

◀ Back ▶ Risk measures ▶ Index

Permanent participation eligibility

$$T_i \in \{0, 1\}, \qquad T_i \text{ remains fixed throughout the path.}$$

- Current holders and reported former holders: $T_i = 1$.
- Among other initial nonholders, $\Pr(T_i = 1) = 0.119$.
- Eligible: annual choice to pay and hold equity. Ineligible: safe asset only.
- Five parameters fitted separately to the same eight financial targets.

A fixed share of nonholders with unknown past participation are permanently excluded.

◀ Back ▶ Financial fit ▶ Dynamic comparison ▶ Index

Financial fit under both access models

Moment	Data	Opportunity	Permanent
Entry: low initial wealth	2.93	3.18	3.21
Entry: middle initial wealth	5.84	5.51	5.49
Entry: high initial wealth	10.36	10.44	10.39
Retention: low initial equity share	66.19	67.75	68.01
Retention: high initial equity share	76.31	74.34	73.83
Next equity share: all initial holders	33.38	34.40	33.81
Wealth change: initial nonholders	3,939	9,941	10,005
Wealth change: initial holders	1,946	511	501

Both models: positive wealth growth, excessive nonholder accumulation.

◀ Back ▶ Permanent eligibility ▶ Index

Entry estimates under permanent eligibility

Entry estimate (pp)	Data	Opportunity net	Permanent net
Upside balance	1.816	0.247	0.171
Upper tail	1.967	0.230	0.155
Lower tail	-1.665	-0.263	-0.188
Variance	0.151	-0.016	-0.016

- Upside balance: pooled 0.211 pp, cell 0.382 pp. A positive association can arise with pooled processes.
- Cells strengthen the association here too. Magnitudes remain well below the data.

Upper- and lower-tail estimates are local derivatives of the same regression.

◀ Back ▶ Permanent eligibility ▶ Index

Participation dynamics under both access models

Untargeted outcome	Data	Opportunity	Permanent
Re-entry: reported former holders (%)	27.16	9.83	57.68
Entry + one later holding (%)	3.48	4.20	3.82
Entry + two later holdings (%)	2.27	3.33	2.24
Projected age-45 ownership (%)	—	31.83	16.31
Windfall response after two years (pp)	—	15.25	3.44

Permanent eligibility: closer fit to longer sustained ownership; excessive re-entry among reported former holders.

◀ Back ▶ Permanent eligibility ▶ Index

Parameter estimates by access model

	Opportunity	Permanent
Risk aversion γ	8.801	8.917
Discount factor β	0.811	0.810
Mean recurring fee κ_R	\$524	\$413
Log-cost standard deviation σ_κ	0.944	1.013
Opportunity q / active share p	0.176	0.119
EIS ψ, fixed	0.5	0.5
Additional entry fee	0	0

Parameters fixed before earnings-risk and dynamic comparisons. No binding bounds.

Estimation criterion and grid refinement

$$Q = Q_E + 0.1Q_H + Q_W = 9.77.$$

- Separate covariance blocks for entry, holder outcomes and wealth changes.
- Wealth covariance: $2,000 regularization. Primary-grid growth buffer: $500 for both groups.
- The optimizer converges. On the finer 201/7 grid: wealth growth of $9,937 for nonholders and $498 for holders.
- Outcomes excluded from estimation: earnings risk, duration, windfalls and life-cycle profiles.

◀ Back ▶ Other numerical diagnostics ▶ Index

Opportunity friction and recurring payments

- Opportunity: a nonholder can reconsider participation.
- Advice, attention and search are possible interpretations, not separately identified.
- Recurring cost paid only when holding equity.
- Offered cost: mean $524, median $335. Participants select lower draws.

Mean recurring fee paid	Real 2018 dollars
Entrants	235
Continuing holders	359

Wealth growth and saving net of returns

$$\Delta A_t = \text{gains on inherited assets} + Y_t - c_t - D_t \kappa_R \xi_t.$$

Nonholder wage tercile	Data growth ($)	Model growth ($)
Low	2,388	3,588
Middle	236	8,705
High	9,195	17,535

Holders have positive wealth growth but negative aggregate saving net of returns.

Asymmetric earnings remain under pooling

- Fixed: education age profiles, group wage levels, initial states, earnings posteriors and calendar returns.
- Set cell growth $B_g(t)$ and departure, scale and tail adjustments to zero; retain asymmetric base shocks.
- Solve choices again at the same financial parameters. The regression still uses the original upside measure; initial differences can still generate an entry association.
- Pooled holder wealth growth: about $483 below zero. Positive growth is imposed in the fitted cell model.

◀ Back ▶ Index

Four-corner accounting by risk measure

c: cell process; p: pooled process; B: entry-regression coefficient.

$B(\text{process}, \text{rule})$: coefficient using the indicated earnings process and decision rule.

$$\Delta_L = \tfrac{1}{2}\{B(c,c) - B(p,c) + B(c,p) - B(p,p)\},$$
$$\Delta_P = \tfrac{1}{2}\{B(c,c) - B(c,p) + B(p,c) - B(p,p)\}.$$

Entry estimate (pp)	Processes	Decisions	Total
Upside balance	+0.274	-0.027	+0.247
Upper tail	+0.302	-0.072	+0.230
Lower tail	-0.245	-0.018	-0.263
Variance	+0.029	-0.045	-0.016

Decision rules dampen the upper-tail contribution and reinforce the negative lower-tail contribution.

One-year accounting for the same observed starting states. Each rule retains the future expectations under which it was solved. Outcomes include cash, work, persistent states and sample selection.

Entry and later ownership: estimates

	One later holding	Two later holdings
Initial nonholders	5,459	1,801
Data upside balance (pp)	0.11	0.77
Data SE	(0.66)	(0.77)
Opportunity net, two seeds	0.19–0.20	0.18–0.22
Permanent net, two seeds	0.14–0.19	0.08–0.13

The smaller samples shared by the data and model give imprecise empirical upside-balance estimates. Simulation-seed ranges are not confidence intervals.

Model fit and remaining discrepancies

Comparison	Model result	Discrepancy
Entry by wealth	Increasing across wealth terciles	Imperfect financial fit
Entry and earnings risk	Opposing net tail signs	Magnitudes well below data
Participation dynamics	Entry rises after gains and windfalls	Sustained levels too high; re-entry too low
Saving and positions	Positive wealth changes for both groups	Excess nonholder accumulation; too few small accounts

Permanent alternative: excessive re-entry; closer sustained-ownership fit.

◀ Back ▶ Alternative dynamics ▶ Index

Earnings asymmetry and joint downturn risk

Study	Relevant finding or comparison
Catherine (2022)	Earnings skewness varies with market conditions. Removing that cyclicality preserves earnings variance, persistence and ordinary earnings–return covariance.
Gálvez & Paz-Pardo (2026)	Compares an untargeted income-risk/risky-share regression in data and models fitted to financial outcomes.
This paper	Earnings upside balance predicts subsequent entry. The model's resources mechanism operates with independent future stock returns.

Upside balance measures upside-minus-downside partial variance, rather than conventional third-moment skewness or dependence on market returns.

Cell/pooled model predictions change the full earnings process, including mean growth. They do not isolate skewness alone.

◀ Back ▶ Comparison scope

Household optimization: states, choices and constraints

Current state	t: age; z: persistent earnings and wage-work states; d: prior stockholding
Resources and fee	m: cash before the fee; $K = \kappa_R \xi$: fee offer; $x = m - jK$: cash after payment
Current choices	$j \in \{0, 1\}$: stockholding; A: saving; α: stock share; consumption $c = x - A$
Next resources	$R(\alpha)$: gross portfolio return; Y': disposable income; primes denote next year

Choose consumption, saving and stocks to maximize lifetime Epstein–Zin utility, subject to

$$c + A + jK = m, \quad c > 0, \quad A \geq 0, \quad 0 \leq \alpha \leq j, \quad \alpha A \geq \$1\, j.$$

β: discount factor; γ: risk aversion; ψ: elasticity of intertemporal substitution.

s_t: survival; q: nonholder access probability; κ_R: mean fee offer; ξ: mean-one multiplier.

▶ Value functions ◀ Back ▶ Access and fees

Where future earnings enter optimal decisions

Let $h = 1 - 1/\psi$ and $\omega = 1 - \gamma$. Process L gives future earnings probabilities.
V^L_{jt}: value conditional on today's stockholding choice j and cash after fees x.

$$V^L_{jt}(z,x) = \max_{(A,\alpha)\in\mathcal{A}_j(x)} \left[(x-A)^h + \beta s_t \left\{\mathbb{E}_L\left[\bar{V}^L_{t+1}(z', j, AR(\alpha)+Y')^\omega \mid z\right]\right\}^{h/\omega}\right]^{1/h}.$$

With access and fee offer $\kappa_R\xi$, choose the better stockholding option:

$$W^L_t(z,m,\xi) = \max\{V^L_{0t}(z,m), V^L_{1t}(z, m-\kappa_R\xi)\}.$$

$\bar{V}^L$: value before access and fee draws; $q_d = d + (1-d)q$.

$$\bar{V}^L_t(z,d,m)^\omega = (1-q_d)V^L_{0t}(z,m)^\omega + q_d\mathbb{E}_\xi[W^L_t(z,m,\xi)^\omega].$$

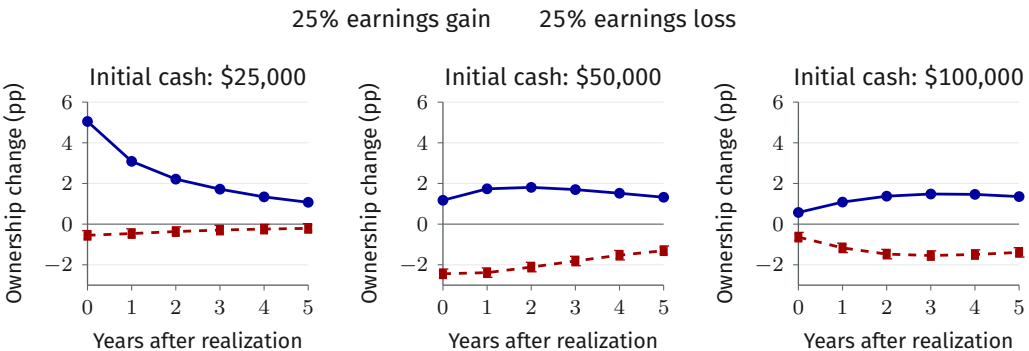
$\mathcal{A}_j(x)$ contains the constraints on the previous slide. At age 84, $\bar{V}^L_{84} = m$.

The earnings process changes expected future utility, and hence today's saving and entry.

◀ Back ▶ Economic interpretation ▶ Index

Response to realized earnings shocks

Age 45: a one-year shock equal to 25% of a fixed reference gross wage



▶ Earnings news at fixed current resources ◀ Back

Realized earnings and earnings news

Experiment	What changes	Interpretation
Realized transitory shock	Current cash; fixed future processes	Gains can finance entry
Upper-tail news	Future distribution; fixed current resources	Participation response can be small
Upside balance news	Future distribution; fixed current resources	Response varies with cash and horizon

Responses to news about upside balance can reverse sign at low cash. More upside need not raise entry.

Numerical precision

- Financial fit, entry projections and four-corner accounting are stable under the specified finer-grid checks.
- Life-cycle grids: 141/5 and 201/7. Some local policies, cash responses and consumption remain grid-sensitive.
- Realized-earnings events: separate 301-node refinement.
- Simulation variance and grid differences measure numerical, not sampling, uncertainty.

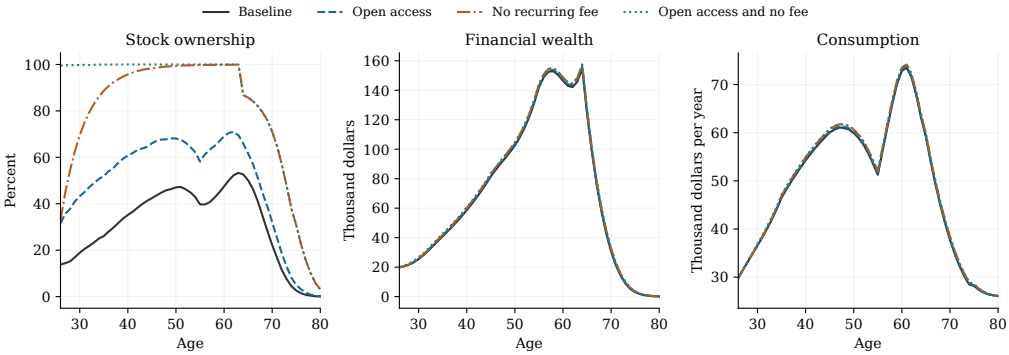
Additional model experiments

Family	Exercises in both specifications
Saving and portfolios	Age profiles, consumption, funded shares, entry and exit
Access frictions	Open access, zero and altered recurring costs
Earnings and retirement	Transitory and persistent risk, income insurance, pensions
Preferences and returns	Risk aversion, patience, EIS, expected returns and volatility
Household responses	Realized earnings, windfalls, fixed-resource news, value of future income

39 scenarios per model, plus conditional policy and valuation diagnostics. Alternative and sensitivity results in the online appendix.

Participation-friction counterfactuals

Remove each friction; reoptimize household choices



◀ Back ▶ Index

SMM targets and weighting

$$\theta = (\gamma, \beta, \kappa_R, \sigma_\kappa, q), \qquad \widehat{\theta} = \arg\min_{\theta} \{Q_E(\theta) + 0.1 Q_H(\theta) + Q_W(\theta)\}.$$

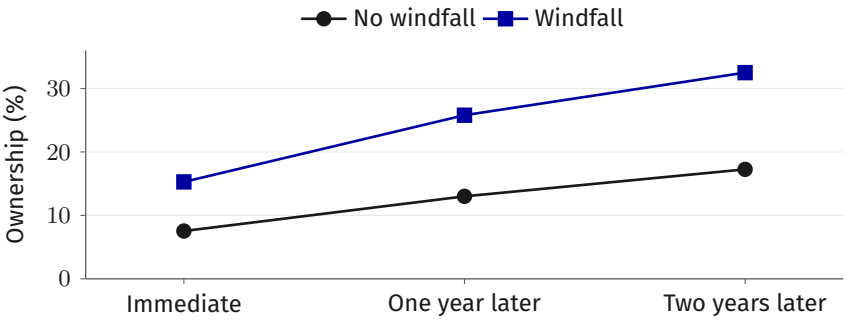
Block	Targets	Economic discipline
Entry (3), Q_E	Three rates by initial wealth	Opportunities and affordability
Holders (3), Q_H	Two retention rates and one next equity share	Continued participation and allocation
Wealth (2), Q_W	Two changes by initial ownership	Saving and resource use

Weight 0.1: entry and wealth growth receive priority over retention and share fit.

◀ Back ▶ Financial fit ▶ All eight targets ▶ Exact criterion

Windfall response

Untargeted experiment: $150,000 to 423 age-44 initial nonholders



Model: $+7.7$ pp immediately; $+15.2$ pp after two years.
Swedish lottery evidence: about $+12$ pp in the winning year (Briggs et al., 2021).

◀ Back ▶ Realized earnings gains ▶ Index

Full entry specification: initial nonholders only

Estimate only among people with no stock or mutual-fund holdings at t.

$$\begin{aligned}\mathbf{1}\{\text{Participates}_{i,t+1}\} = \alpha &+ \beta_E \widetilde{C}_{c(i,t)} + \gamma_1 \widetilde{V}_{c(i,t)} + \gamma_2 \widetilde{\sqrt{V}}_{c(i,t)} \\ &+ \gamma_3 \widetilde{\mu}_{c(i,t)} + \gamma_4 \left(\widetilde{\mu}_{c(i,t)}\right)^2 \\ &+ \delta_1 \log\Big(\max\{A_{it}, 1\}\Big) + \delta_2 \log\Big(\max\{\bar{y}^{(1)}_{it} + 1, 1\}\Big) \\ &+ \delta_3 \mathbf{1}\Big\{\min_{m \in \mathcal{M}_{it}} y^{(1)}_{itm} \le 0\Big\} + \delta_4 a_{it} + \delta_5 a_{it}^2 + \delta_6 \text{Male}_{it} \\ &+ \eta_{j(i,t)} + \kappa_{e(i,t)} + \lambda_{b(i,t)} + \tau_{t+1} + u_{i,t+1}.\end{aligned}$$

Earnings measures: C upside balance; V variance; μ mean adjusted growth. Tildes denote standardization; c is the baseline earnings cell.

Resources: A gross assets; $y^{(1)}$ first-listed-job earnings; $\bar{y}^{(1)}$ its mean over eligible months $\mathcal{M}$. The indicator records any month with nonpositive earnings. a is age.

Fixed effects: j industry, e education, b age band, plus outcome year; no person FE. Participation is positive ownership in any eligible month of the year.

◀ Back ▶ Measurement versus controls ▶ Inference

Does the entry association depend on the earnings measure?

	(1)	(2)	(3)	(4)
	Wages	Total earnings	Total earnings	Total earnings
	zeros included	zeros included	positive only	positive only
Growth measure	Arc	Arc	Real log	Arc
Upside balance C	1.601^{***}	1.042^{**}	0.875^{***}	0.695^{***}
	(0.506)	(0.405)	(0.279)	(0.248)
Split-half reliability of C	0.73	0.62	0.28	0.24
Age profile removed in CPS	Yes	Yes	Yes	Yes
V, $\sqrt{V}$, mean and mean^2	Yes	Yes	Yes	Yes
Resources and demographics	Yes	Yes	Yes	Yes
Industry, education, age band, year effects	Yes	Yes	Yes	Yes

Every column: 27,419 nonholder transitions; mean entry 8.82%; 58 earnings cells and 30 clusters.

Reliability asks whether two random halves of CPS workers give similar cell measures; values use the Spearman–Brown adjustment. Removing the age profile constructs the CPS measure; it is not a second cubic control in SIPP.

◀ Back ▶ Variance coefficients ▶ Controls ▶ Inference

Additional specification results

Specification	Upside balance (pp)	SE
Baseline: valid shares at both dates	1.65**	0.60
Industry × year; education × year	1.66**	0.62
Industry × education effects	1.86**	0.88
Reported, self-reported, stable households	2.56***	0.56
Controls for realized earnings	1.57**	0.59

The first three rows use 24,235 initial nonholders with positive broad and nonretirement financial assets and valid shares at both dates. Reporting screens and realized-earnings controls use their own samples.

▶ Person FE and sorting ▶ Exact samples

Variance parameters and earnings asymmetry can coexist

Using variance parameters does not make an earnings process symmetric.

For example, normal log wages with positive variance imply skewed wage levels:

$$\log W \sim N(m, \sigma^2) \quad \Longrightarrow \quad \text{Skew}(W) = (e^{\sigma^2} + 2)\sqrt{e^{\sigma^2} - 1} > 0.$$

Here even changing a variance parameter changes wage-level skewness. This example concerns levels; our empirical measures describe wage growth.

Jump and mixture models allow additional shape variation through jump sizes, component means and probabilities. Variances and tail balance depend jointly on the full law.

Our question: does the measured upside–downside balance predict entry beyond mean growth and total variance?

We use existing asymmetric earnings-model ingredients to examine that relationship.

◀ Back ▶ Our earnings process ▶ Growth skewness and upside balance

Which comparisons are performed?

Comparison	Scope
Upside balance and skewness	Closely related. Joint regressions do not establish incremental information beyond skewness.
Cell and pooled earnings	Same model and financial parameters; optimal rules recomputed. This is the main earnings-environment comparison.
Opportunity and permanent access	Separately fitted to the same financial targets; different discrepancies in participation dynamics
Published structural models	Mechanism comparison. No common-data re-estimation of entire published models.

◀ Back ▶ Skewness horse race ▶ Access models

Upside balance and skewness in total earnings

Total earnings, including zeros	Result
Upside balance coefficient (pp)	3.782
Two-survey standard error (pp)	2.261
Upside balance p-value	0.100
Two-test adjusted q-value	0.200
Joint test: both slopes zero	$p = 0.038$
Joint test: broader dependence sensitivity	$p = 0.145$

Total earnings, including zeros; adds zero-wage probability. Separate two-survey inference.

This is not the headline wage regression. The evidence does not establish an incremental association beyond skewness.

Measurement adjustment and regression controls

Step	Purpose
CPS earnings construction	Remove an industry–education-specific age profile from growth, then calculate within-cell distributional moments.
SIPP entry regression	Condition participation on individual resources, age and sex, plus industry, education, age-band and year fixed effects.

The CPS age cubic is a measurement step, not a repeated SIPP control. Individual age and group effects in SIPP adjust the participation comparison.

Sex is an individual participation control; the headline risk cells are not sex-specific.

Separate industry, education and age-band effects do not absorb every industry–education–age interaction.

◀ Back ▶ Construction ▶ Full regression

The coefficient's unit

$$\widetilde{C}_c = \frac{C_c - \bar{C}_{\text{CPS}}}{s_{C,\text{CPS}}}, \qquad s_{C,\text{CPS}} \simeq 0.034.$$

A one-unit increase in $\widetilde{C}$ means a 0.034 increase in raw upside balance.

The fitted coefficient is 0.01601 in probability units, or 1.601 percentage points.

Rescaling by the variation left after controls gives 0.590 pp per residual SD (0.369 of the original SD). The regression coefficient is unchanged.

This unit describes differences between earnings environments. It is not a 1% earnings rise or an individual windfall.

◀ Back ▶ Tail units ▶ Illustration

Samples and outcomes across surveys

Question	Observations	Reason
Main SIPP entry	27,419	Initial nonholders with required annual outcomes and controls
Allocation checks	24,235	Initial nonholders with positive financial denominators and valid shares at both dates
Entry plus later holdings	9,236 / 3,254	One / two additional complete followups
Model financial fit	21,188	Complete initial states for the model, including initial holders
Model entry regression	17,606	Initial nonholders within the model sample
SCE reallocation / share	2,136 / 3,332	Different outcome availability and eligibility

Model and data coefficients are compared on the same 17,606 initial-nonholder observations. The 27,419-observation headline uses the broader entry sample.

◀ Back ▶ Additional samples

Initial nonholders and subsequent ownership sequences

Initial nonholders with observed ownership in every required follow-up year.

Four-year sequence	Entry	Entry + two later holdings
$0, 1, 1, 1$	1	1
$0, 1, 0, 1$	1	0
$0, 0, 1, 1$	0	0
$1, 0, 1, 1$	Outside the initial-nonholder sample	

Each year defines a new baseline. A later $0, 1, 1$ window can enter the shorter-horizon cohort.

Missing required follow-up excludes the observation. Ownership means any stockholding during that year.

Retention among entrants is a separate outcome; its estimates are imprecise.

◀ Back ▶ Exit ▶ Joint outcomes

Regressions of simulated entry probabilities

Y: observed entry indicators; X: regressors; W: survey-weight matrix.

$\widehat{\beta}$: weighted regression coefficients; p: predicted entry probabilities.

$$\widehat{\beta}(Y) = (X'WX)^{-1}X'WY.$$

If $p = \mathbb{E}[Y \mid X]$, then

$$\mathbb{E}[\widehat{\beta}(Y) \mid X] = \widehat{\beta}(p).$$

Averaging binary realizations into probabilities removes simulation noise; it does not mechanically reduce the expected regression coefficient.

Mean entry in the matched sample: model 6.375%, data 6.375%. None of the 17,606 predicted probabilities is exactly zero.

The shortfall concerns how probabilities vary with earnings upside balance. Model probabilities need not equal the true conditional expectation.

◀ Back ▶ Coefficient comparison

Alternative specifications do not close the magnitude gap

Specification	Upside balance (pp)	Nonholder wealth error	Holder wealth error
Current cell model	0.241	Reference	Reference
Cash-dependent opportunities	0.337	+$157	+$11
$50 recurring cost + setup	0.267	+$302	+$10
$100 recurring cost + setup	0.253	+$362	+$11
Matched empirical coefficient	1.816		

Wealth columns: change in each group's absolute wealth-growth error; positive is worse. Same 141-node resolution. Both groups also worsen on the finer grid.

Low-fee versions predict retention above 96%. None closes the entry-coefficient gap without worsening the wealth fit.

◀ Back ▶ Permanent alternative

The earnings process: states and wage levels

$$F^{L}_{g,t}(\eta_{t+1}, n_{t+1}, W_{t+1} \mid \eta_t, n_t), \qquad L \in \{\text{cell}, \text{pooled}\}.$$
$$W_{gt} = \mathbf{1}\{n_t = 0\} \exp\{f_h(t) + \ell_g + B_g(t) + \eta_t + \varepsilon_t\}.$$

State / component	Meaning
n_t	0 with wage work, 1 without wage work
$h,\ g$	Education type, industry–education group
$f_h(t),\ \ell_g$	Education age profile and group wage level
$B_g(t),\ \eta_t,\ \varepsilon_t$	Cell growth profile, persistent state, transitory shock

Gross wages W_{gt} enter disposable resources Y_t after taxes and employee contributions. Other resources can remain when wages are zero.

◀ Back ▶ Cell coefficients ▶ Career transitions ▶ Cell and pooled

The cell adjustment vector and its joint targets

Coefficient	What changes
a_c^{exit}	Wage-work departure hazard
b_c	Deterministic age-growth profile
s_c	Transitory and reporting-error scale
a_c	Transitory probabilities and persistent jump hazards

$$B_g(t) = \begin{cases} b_{g0}(t-35), & t < 45, \\ 10b_{g0} + b_{g1}(t-45), & t \geq 45. \end{cases}$$

Fit the four coefficients jointly to the zero-wage share, mean arc growth, and upper/lower partial variances.

Pooling sets $(b_c, s_c, a_c, a_c^{\text{exit}}) = (0, 0, 0, 0)$ in every cell. Then $B_g = 0$, the scale and departure multipliers equal 1, and the tail weights return to baseline. Shared earnings shocks remain.

◀ Back ▶ Persistent transitions ▶ Temporary shocks

Persistent earnings and wage-work transitions

$$\eta_{t+1} = \rho_h \eta_t + \zeta_{t+1} + J_h^+ I_{t+1}^+ - J_{h,b}^- I_{t+1}^- - m_{g,b}, \quad \zeta_{t+1} \sim N(0, s_{0h}^2).$$

ρ_h: persistence. $m_{g,b}$: centering drift. Promotion, loss and no jump are mutually exclusive outcomes.

$$\widetilde{p}_{c,b}^{\pm} = \min\{\widehat{p}_{c,b}^{\pm}, .45\}, \qquad p_{c,b}^{\pm}(\eta) = \min\{\widetilde{p}_{c,b}^{\pm} e^{\pm k_h^{\pm} \eta / \sigma_{\eta,h}}, .45\}.$$

$\widehat{p}^{\pm}$ are the tilted probabilities before caps, defined on the linked normalization slide.

$$\Pr(n_{t+1} = 1 \mid n_t = 0, \eta_t) = \min\{\delta_{h,b} e^{a_c^{\text{exit}}} e^{-\phi_h \eta_t / \bar{\eta}_h}, .6\},$$

$$\Pr(n_{t+1} = 0 \mid n_t = 1) = \min\{r_{h,b}^0 e^{r_h}, .95\}.$$

$n = 0$: wage work. $n = 1$: no wage work. Earnings states evolve in both states. Wage-work return r_h differs from investment opportunity q.

◀ Back ▶ Tilt and normalization ▶ Parameters

Normalization of persistent earnings shocks

Before caps and state dependence, cell c tilts promotion/loss probabilities:

$$\widehat{p}^{+}_{c,b} = \chi_{c,b} p^{+}_{h,b} e^{a_c}, \qquad \widehat{p}^{-}_{c,b} = \chi_{c,b} p^{-}_{h,b} e^{-a_c},$$

$$\chi_{c,b} = \frac{p^{+}_{h,b}(J^{+}_{h})^2 + p^{-}_{h,b}(J^{-}_{h,b})^2}{p^{+}_{h,b} e^{a_c}(J^{+}_{h})^2 + p^{-}_{h,b} e^{-a_c}(J^{-}_{h,b})^2}.$$

$$\widehat{p}^{+}_{c,b}(J^{+}_{h})^2 + \widehat{p}^{-}_{c,b}(J^{-}_{h,b})^2 = p^{+}_{h,b}(J^{+}_{h})^2 + p^{-}_{h,b}(J^{-}_{h,b})^2.$$

This fixes a raw jump second moment before later adjustments.

Probability caps and state dependence follow. Centering, employment transitions and temporary shocks also matter for total earnings variance and growth.

◀ Back ▶ Complete transition ▶ Temporary component

Transitory earnings and reporting error

Transitory earnings

$$\varepsilon^0 \sim (1-p_2)N(0,s_1^2) + p_2 N(m_2, s_2^2).$$

Ten raw nodes e_j with probabilities p_{0j}. Cell tilt and scale give

$$p_{cj} \propto p_{0j} \exp\{a_c \operatorname{sgn}(x_j) x_j^2\}, \quad x_j = \operatorname{clip}\big((e_j - E_0 e)/sd_0(e), -4, 4\big),$$

$$\varepsilon_{cj} = e^{s_c} e_j - \log \sum_k p_{ck} e^{e^{s_c} e_k} + \log \sum_k p_{0k} e^{e_k}.$$

The location adjustment preserves $E[e^{\varepsilon}]$, the baseline wage multiplier. It does not fix variance.

Reporting error

$$\log W_t^{\text{obs}} = \log W_t + u_t \quad (W_t > 0), \qquad u_t^0 \sim (1-p_u)N(0,t_1^2) + p_u N(0,t_2^2).$$

Cell scale also multiplies reporting-error nodes by e^{s_c}. Reporting error changes measurement, not household resources.

◀ Back ▶ Shock parameters ▶ Cell coefficients

Earnings parameters: career dynamics

Parameter	Noncollege	College
Persistence ρ_h	0.946	0.944
Small innovation SD s_{0h}	0.024	0.037
Promotion size J_h^+	0.328	0.522
Loss size J_h^-, ages 25–54	0.742	0.419
Loss size J_h^-, ages 55–64	1.150	1.346
Promotion state sensitivity k_h^+	0.248	0.140
Loss state sensitivity k_h^-	0.148	0.931
Age-band order: 25–34, 35–44, 45–54, 55–64		
Base promotion probabilities $p_{h,b}^+$	.117, .039, .013, .046	.082, .019, .009, .021
Base loss probabilities $p_{h,b}^-$	.036, .032, .023, .035	.010, .029, .030, .043
$f_h(t) = a_{0h} + a_{1h}x + a_{2h}x^2 + a_{3h}x^3, \quad x = (t-45)/20$		
a_{0h}, a_{1h}	10.502, .202	11.153, .118
a_{2h}, a_{3h}	$-.239, -.004$	$-.361, .087$

Jump sizes in log points; annual probabilities before cell and state adjustments.

◀ Back ▶ Other parameters ▶ Main earnings slide

Earnings parameters: shocks and wage work

Parameter	Noncollege	College
Transitory large-component probability p_2	.044	.031
Transitory small-component SD s_1	.060	.101
Transitory large-component mean m_2	$-.339$	$-.349$
Transitory large-component SD s_2	.627	.620
Reporting large-component probability p_u	.230	.231
Reporting small-component SD t_1	.164	.182
Reporting large-component SD t_2	.913	.819
Departure state sensitivity ϕ_h	1.463	1.435
Wage-work return log shift r_h	1.926	1.996
Age-band order: 25–34, 35–44, 45–54, 55–64		
Base departure probabilities $\delta_{h,b}$	.074, .068, .057, .073	.039, .030, .041, .069
Return inputs $r^0_{h,b}$	.244, .170, .084, .047	.924, .292, .263, .094

Log-point shocks. Return inputs precede the fitted shift and probability cap.

◀ Back ▶ Career parameters ▶ Process definition ▶ Main

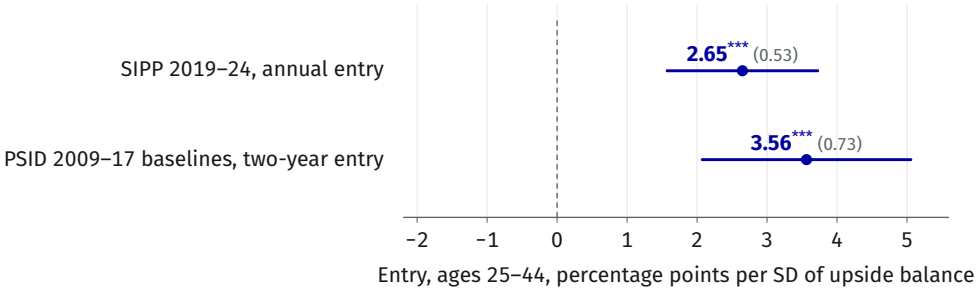
Predetermined measurement of earnings risk

- CPS earnings changes are measured before the SIPP investment outcomes.
- A current shock cannot directly enter both the historical risk measure and the current portfolio report.
- Persistent group conditions, worker sorting and correlated shocks can still connect the two periods.

The timing reduces a source of contemporaneous correlation. It does not establish causal assignment of earnings risk.

◀ Back ▶ Group controls

Entry at ages 25–44



▶ Period gap and all-age results

Earnings fit and participation predictions

1. Fit wages and wage-work transitions to earnings data.
2. Financial targets discipline saving, participation costs and access.
3. Neither step targets how entry varies with earnings upside balance.

The model must convert those earnings distributions into the observed entry response through resources and choices.

Matching the earnings moments alone does not guarantee the sign or size of that response.

◀ Back ▶ Fitting targets ▶ Mechanisms

Published magnitude benchmarks

Comparisons for a one-SD difference in each study's predictor.

Study / population	Predictor and outcome	Magnitude
This paper: US nonholders	Upside balance at fixed variance; entry over one year	$+1.601$ pp; $+18.16\%$ of mean entry
Kaustia–Knüpfer (2012): Finland	Local-peer monthly returns; entry the following month	$+9$–13% relative
d'Astous–Shore (2024): Danish university applicants	Permanent earnings volatility, 11% to 14% annually; current ownership	-6.5 pp: 28% to 21.5% (-23%)

Entry is a transition among nonholders; ownership is a stock. Predictors, horizons and populations differ.

Kaustia–Knüpfer use peer-return panel regressions; d'Astous–Shore use university-admission discontinuities. These are magnitude benchmarks, not estimates of the same relationship.

Sources: Kaustia & Knüpfer (JFE, 2012), introduction; d'Astous & Shore (JFE, 2024), introduction.

◀ Back ▶ Normalization ▶ Index

Related work in progress

van den Akker, de Bresser, Teppa & Trucchi (2026)

Subjective Expectations and Household Portfolio Choices

- Subjective expectations linked to administrative portfolios.
- Abstract: upward income risk predicts investment; entry patterns are concentrated among younger households.

Here: measured group earnings upside balance and entry, controlling for total variance and resources.

Conference presentation listed in the IAAE 2026 program. The comparison is based on the conference abstract.

Two-period illustration: the optimization problem

Choose consumption c_0, safe saving s, stock investment x and entry $d \in \{0, 1\}$:

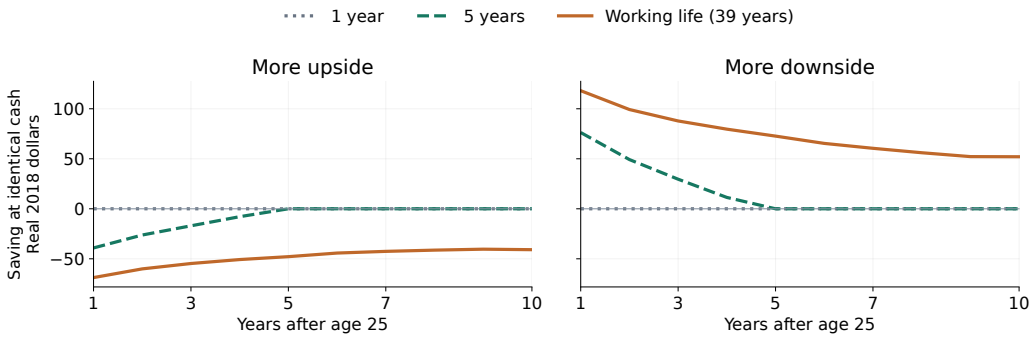
$$\max_{s,x,d} \log c_0 + \mathbb{E} \log c_1, \qquad c_0 = m - s - x - 1.5d, \qquad c_1 = Y_1 + s + Rx.$$

- Same log utility and discount factor $\beta = 1$. No borrowing or shorting: $s, x \geq 0$, with $x = 0$ if $d = 0$ and $c_0 > 0$.
- Safe gross return is 1. Stock gross return R is 0.5 or 1.7, each with probability one half, independent of earnings.
- Y_1 follows U or D. In the realized-resource panel, $m = Y_0$ with zero inherited assets; Y_0 and Y_1 are independent draws from the same process.
- Financial saving is $s + x$: it excludes entry fees and subsequent returns. All amounts are in thousands of dollars.

Optimize with and without entry, then choose the higher value. These are teaching assumptions, not estimated parameters of the life-cycle model.

◀ Back ▶ Expectations ▶ Resources

Saving responses at fixed current resources



In this illustration, lasting upside lowers saving even without current stock investment. At the same cash: $69 less in year 1 and $41 less in year 10. The one-year response is zero because future earnings are already baseline at the first new choice.

◀ Back ▶ Entry at the same cash ▶ All diagnostics

Full responses and accounting comparisons

How do later first-entry rates change? Annual first entry

Do expectations change choices at the same current state? Entry decisions Saving

Dynamic paths: saving can change later cash

Duration	Entry	Wealth	Budget
One year	Decomposition	Decomposition	Accounting
Five years	Decomposition	Decomposition	Accounting
Working life	Decomposition	Decomposition	Accounting

All comparisons use the same initial nonholders and fixed parameters. The shocks illustrate mechanisms. Earnings-risk coefficients are untargeted.

◀ Back ▶ Experiment design

Lasting upside: fewer first-time entrants in later years



Fewer people first enter from year 3 onward under lasting upside.
The early entry lead shrinks. Cumulative first entry still counts those who entered earlier.

◀ Back ▶ Cumulative entry ▶ Same-cash decisions

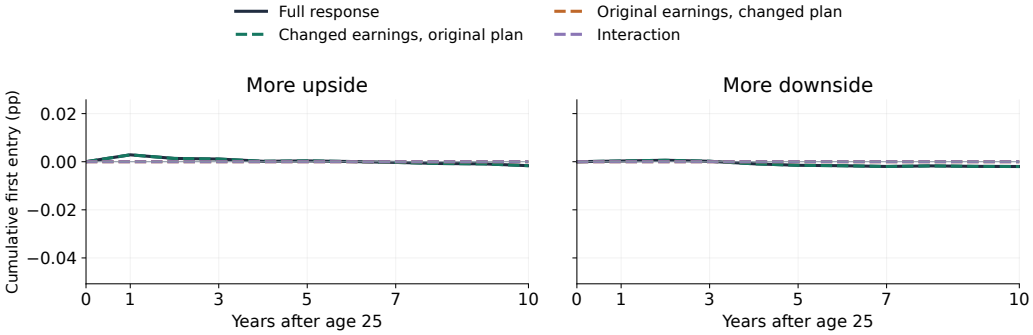
Illustration: lasting upside encourages entry at the same cash



The incentive to enter remains in year 10 at the same current state.
Compare current nonholders at the same resources and other states on baseline paths. This includes re-entry. These choices do not change later simulated states.

◀ Back ▶ Same-cash saving ▶ All diagnostics

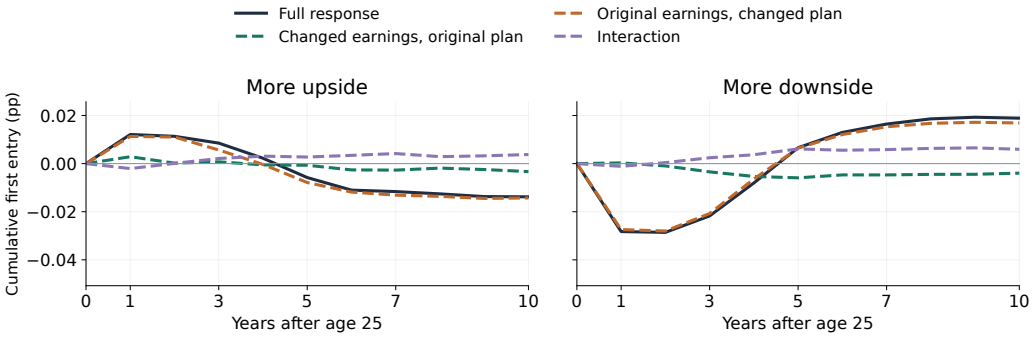
One year: earnings paths and entry



Only realized resources change at the first new decision.
The one-year change has ended for all future earnings. Rule-only and interaction responses are exactly zero. Tiny entry signs are grid-sensitive.

◀ Back ▶ Entry ▶ Wealth ▶ Budget

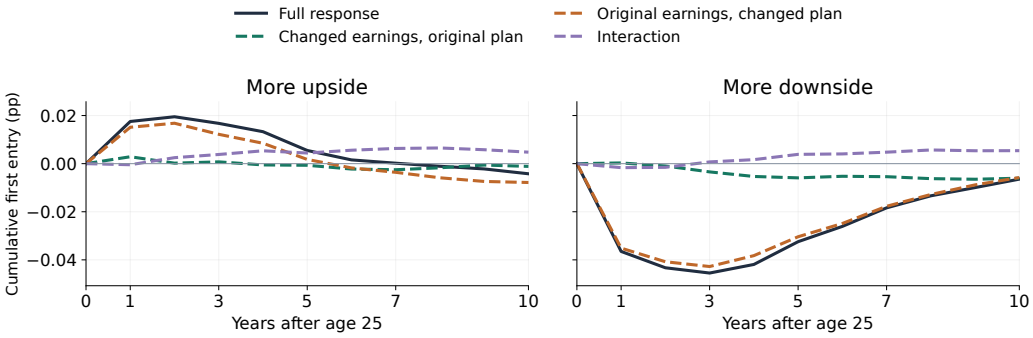
Five years: earnings paths and entry



Different saving and portfolio decisions account for most of the later reversal in cumulative entry. The decision-rule comparison includes saving and its effect on later resources. Current cash is therefore not held fixed along these paths.

◀ Back ▶ Entry ▶ Wealth ▶ Budget

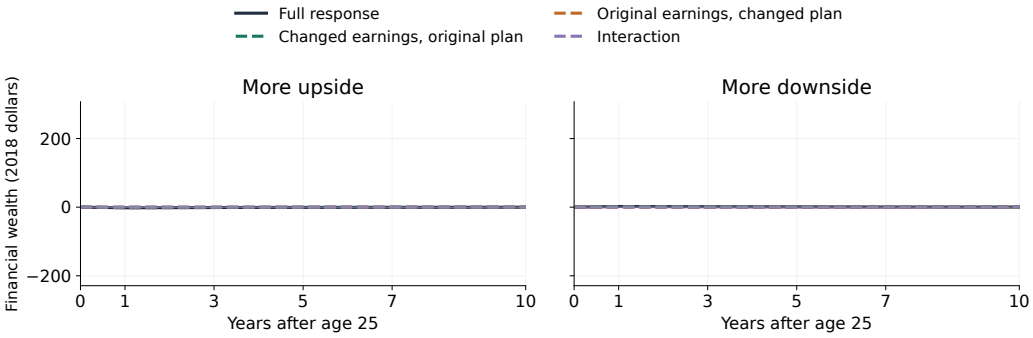
Working life: earnings paths and entry



The full cumulative-entry response ends close to zero.
Changes in earnings and decisions interact. Decisions include saving and investment, which affect later resources.

◀ Back ▶ Entry ▶ Wealth ▶ Budget

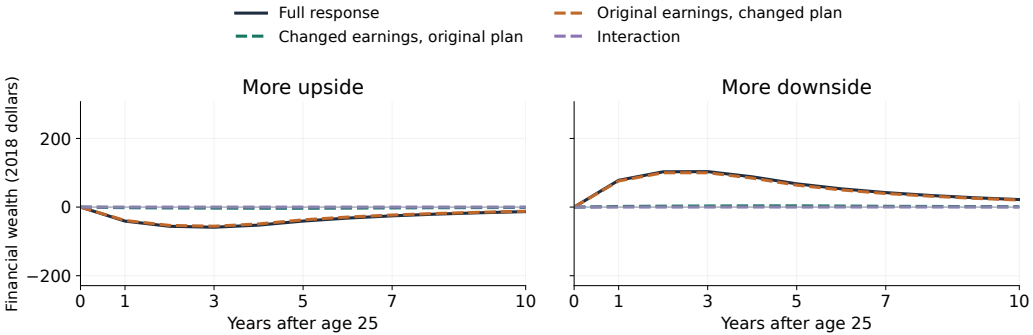
One year: earnings paths and wealth



The small wealth effect comes through realized resources.
Future earnings prospects are baseline at the first new choice. Households use the original rule at their changed cash.

◀ Back ▶ Entry ▶ Wealth ▶ Budget

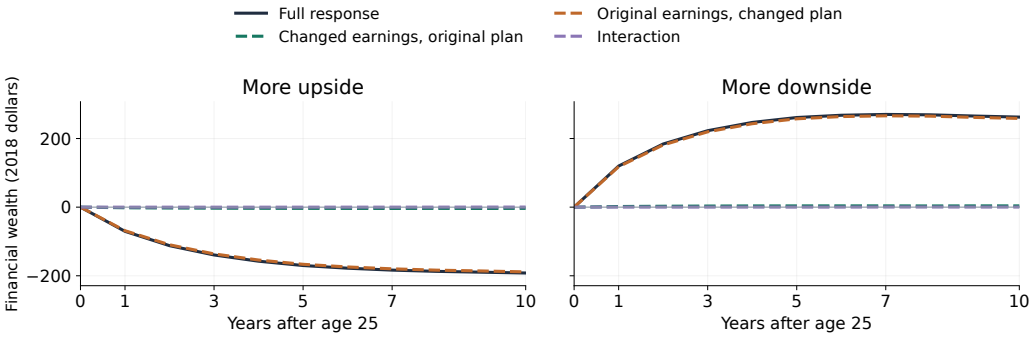
Five years: earnings paths and wealth



Earlier choices leave a wealth gap after the regime ends.
Rules coincide at the same state from year 5. The financial resources reached along each path can still differ.

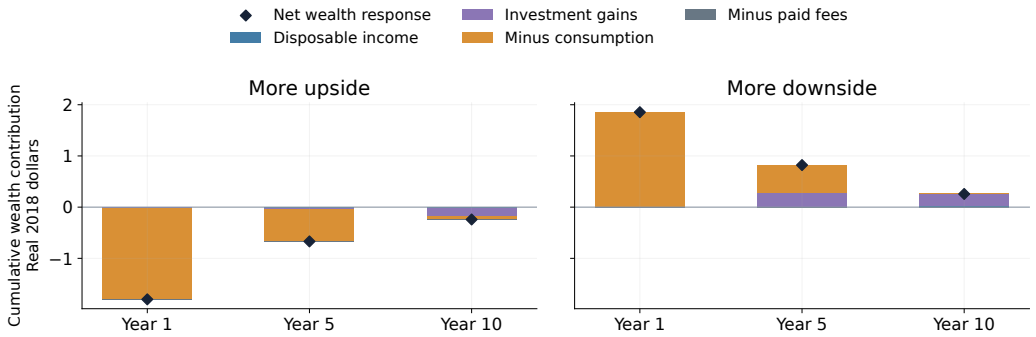
◀ Back ▶ Entry ▶ Wealth ▶ Budget

Working life: earnings paths and wealth



Changes in saving and portfolio decisions account for most of the persistent wealth gap. At year 10 under upside, the decision-rule contribution is about -$189. Changed earnings under the original decision rule contribute about -$4.

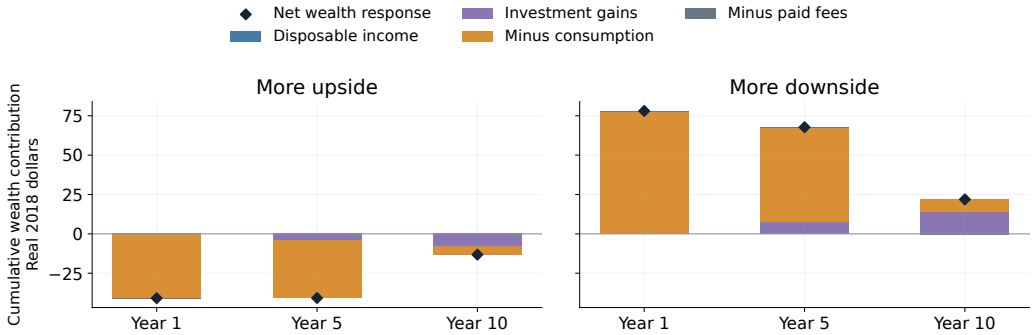
One year: where does the wealth difference come from?



A one-year change produces a small budget response.
After current earnings arrive, future prospects are already baseline. The current income draw changes resources, consumption and saving.

◀ Back ▶ Wealth decomposition ▶ Saving at the same cash

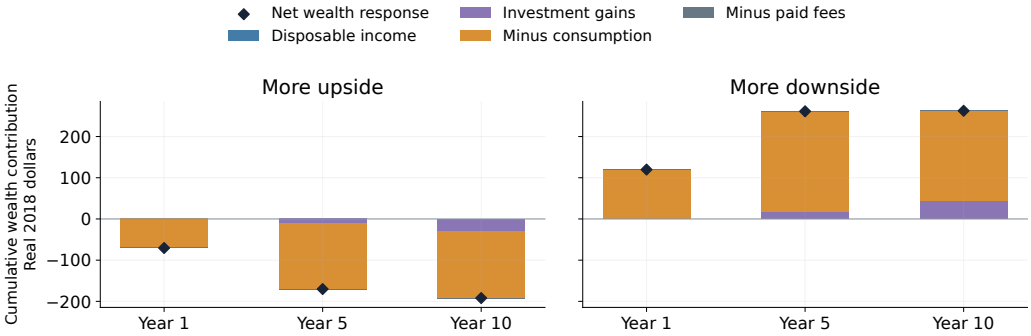
Five years: where does the wealth difference come from?



Consumption explains most of the initial wealth response.
As the temporary change ends, spending responses diminish. Earlier saving and investment choices still affect wealth.

◀ Back ▶ Wealth decomposition ▶ Saving at the same cash

Working life: where does the wealth difference come from?



Higher consumption accounts for most of the upside wealth decline.
At year 10: about -$163 from consumption, -$28 from investment gains and -$0.18 from fees.

◀ Back ▶ Wealth decomposition ▶ Saving at the same cash